The Impact Frontier

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Abstract

How much can a sustained, marginal portfolio tilt by an investor change firms' capital and the external outcomes those firms produce? This paper develops an equilibrium theory and empirical calibration of the capital-allocation channel of investor impact. The model nests the CAPM, incorporates investor heterogeneity through a non-market-risk term, and aligns firm capital demand with q-theory. A contribution multiplier matrix converts an investor's supply-schedule shift into equilibrium changes in firm capital. Combined with outcome accounting, this matrix produces impact returns: expected outcome changes per dollar of position value. The Impact Frontier is the Pareto frontier between certainty-equivalent returns and portfolio impact returns.

Calibrated to U.S. public equities with five-year return and capital-growth windows, the estimated contribution multipliers exhibit substantial cross-sectional heterogeneity. Tilts that account for this heterogeneity yield impact returns close to the frontier, while tilts based only on raw firm outcomes can fall far below, reflecting the gap between firm-level attribution and investor contribution in equilibrium.

Keywords: Asset pricing, capital allocation, capital budgeting, investor impact, general equilibrium, portfolio choice, sustainable investing.

JEL Classification: C36, G10, G11, G12, G31, Q56.

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1 Introduction

Many investors seek to improve social or environmental outcomes by holding more of some firms and less of others. Whether such portfolio choices change real activity is not obvious. How much can a sustained, marginal portfolio tilt by an investor change firms' capital and the external outcomes those firms produce? The answer depends on both sides of the capital market. Investors supply capital on terms that depend on risk, mandates, and preferences. Firms demand capital when investment opportunities are attractive on the offered terms. A tilt shifts one investor's capital-supply curve; market clearing then determines how capital is reallocated across firms and how that changes the external outcomes.

This paper studies that mechanism as the capital-allocation channel of investor impact: the expected change in an aggregate real outcome caused by investor action (Kölbel et al. 2020; Brest et al. 2018; Heeb et al. 2023; Yasuda 2025). Rather than treating additionality as a binary label (Carter et al. 2021), the model measures investor contribution on the margin: how much of a supply shift becomes issuer capital and how much outcome change follows. Engagement, governance, philanthropy, policy advocacy, and other channels can also generate investor impact. The question here is channel-specific: how much outcome change per unit investor wealth can a marginal portfolio tilt generate?

Standard asset pricing theories usually model neither firm capital quantities nor effects on real outcomes, but applied to the capital-allocation channel they would push toward opposite extremes. On the investor side, in the CAPM, diversified investors absorb firm-specific risk, so the supply of capital is highly elastic and tilts have little effect (Markowitz 1952; Sharpe 1964; Berk and Van Binsbergen 2025). Recent demand-system estimates instead imply finite and often low investor-side price elasticities, alongside sometimes significant demand for ESG characteristics (Koijen and Yogo 2019; Gabaix and Koijen 2021; Koijen et al. 2024; Haddad et al. 2025; Li et al. 2025; Van der Beck 2026). Recent theory cautions that cross-asset complementarities and equilibrium price spillovers can bias demand-system elasticities (Fuchs et al. 2025). Portfolio-choice models imply larger, but still finite, stock-level elasticities (Davis et al. 2025). Related evidence from fiduciary-duty shocks, dividend-flow price pressure, and price-agnostic demand is also consistent with investor-side inelasticity

(Cassella et al. 2026; Hartzmark and Solomon 2025; Hartzmark and Sussman 2026).

On the firm side, endowment-economy models assume the firm side is inelastic (Lucas 1978), while production-economy models can generate highly elastic capital demand when production is near constant returns to scale (Favilukis et al. 2025). Between these benchmarks, Choi et al. (2025) estimate a dynamic, quarterly corporate-finance model with finite issuer (and investor) elasticities, providing a useful comparison for the estimates reported here.

This paper estimates both investor-supply and issuer-demand slopes rather than imposing either limiting case. The model is a first-order equilibrium system: tractable enough to estimate, but detailed enough to preserve the cross-sectional heterogeneity that matters for portfolio theory in general equilibrium (Cochrane 2022). The paper makes six contributions, moving from the equilibrium model to slope estimation and then to calibrated Impact Frontiers that quantify the return-impact tradeoffs implied by the estimated system.

First, it develops a supply-and-demand system for risky assets, motivated by Betermier et al. (2026). The investor-supply equation includes the standard market risk term, nesting the CAPM, plus a non-market-risk term that reflects investor heterogeneity. This heterogeneity may arise from information costs, short-sale constraints, principal-agent frictions, trading horizons, and other limits to investor participation (Merton 1987; Hong and Sraer 2016; Ewens et al. 2013; Balasubramaniam et al. 2023; Starks et al. 2026; Dello Preite et al. 2025). The issuer-demand equation aligns with q-theory and production-based asset-pricing: issuer capital demand responds to the cost of capital, with heterogeneity coming from existing issuer scale and risk (Zhang 2017; Belo et al. 2013; Gonçalves et al. 2020).

Second, it derives the equilibrium response to a marginal investor supply shift. In the one-asset case, the direct contribution multiplier is the share of the shift that becomes firm capital. For multiple assets, the contribution multiplier matrix converts portfolio tilts into capital changes across firms. In equilibrium, aggregate issuer-capital changes are matched by changes in the supply of the risk-free asset.

Third, the model adds an aggregate external outcome. Each firm has a net outcome intensity: its outcome intensity per unit of issuer capital minus the outcome intensity assigned to the risk-free asset. This netting makes the frontier invariant to arbitrary level shifts in gross outcome measures, including for subjective scores without a natural zero. Applying the contribution multiplier matrix to the net outcome intensities and dividing by prices yields impact returns: expected outcome changes per dollar of position value. The studied tilts are sustained changes in an investor's capital-supply schedule over the model horizon, so the resulting impact returns are marginal effects of positions held over that horizon rather than temporary reallocations.

Fourth, the paper derives the Impact Frontier, the Pareto frontier between certainty-equivalent (CE) return and portfolio impact return. It also decomposes frontier strength into interpretable quantities and characterizes worst-case performance for different tilt rules.

Fifth, it estimates the marginal supply and demand slopes that govern the capital-allocation channel in U.S. public equities using five-year return and capital-growth windows. The empirical specification is parsimonious: a market-risk supply slope, a non-market-risk supply slope, and one issuer-demand slope summarize the estimated margins. Estimating both sides in comparable units in one simultaneous system lets the slopes jointly determine the contribution multiplier matrix. The market-risk term alone implies finite investor-supply elasticity: more elastic than the most inelastic demand-system estimates, but far from the frictionless CAPM limit. The estimated issuer-demand and investor-supply elasticities are of the same order of magnitude, so the contribution multipliers vary substantially between zero and one rather than clustering near either limit. The non-market-risk term accounts for nearly half of the market-portfolio supply slope and substantially raises the median direct contribution multiplier, but the resulting frontiers still depend on the full system.

Sixth, using ESG scores as illustrative outcome-intensity proxies, the paper constructs calibrated Impact Frontiers and compares them with simple portfolio rules. Frontier strength and the performance of simple rules depend on the supply-and-demand system, the outcome-intensity distribution, and the investor's risk preferences. Tilts that account for the direct contribution multipliers are often close to the Impact Frontier in the full cross-section; tilts based only on net outcome intensities can fall far below it, especially within narrower investment universes.

Portfolio tilts are analyzed as supply-schedule shifts rather than as holdings choices. Choosing holdings while leaving issuer capital fixed would rule out the capital-allocation channel by construction. Instead, market clearing determines the first-order changes in prices, issuer capital, and all investors' positions. Marginality makes second-order terms negligible and holds the surrounding schedules fixed; it does not set the capital response to zero. The Impact Frontier compares the CE-return cost of the tilt with its portfolio impact return, without assigning a value to the external outcome or modeling the absolute level of expected returns.

Evidence from private equity, bank exit policies, and equity-dependent firms shows that financing conditions can affect real investment in some settings (Baker et al. 2003; Cole et al. 2025; Green and Vallée 2025). Studies of SRI funds, ESG-rating shocks, and expert beliefs suggest more limited real effects for typical sustainable equity funds (Heath et al. 2023; Berg et al. 2024; Heeb et al. 2025). Together, this evidence motivates an equilibrium model that separates investor preferences from real effects, distinguishes average from marginal effects, and traces how tilts affect real outcomes.

Beyond the empirical evidence, theoretical work in sustainable finance studies how investor preferences, impact motives, ESG information, fund mandates, and sustainability standards affect firm behavior, investment, asset prices, and the cost of capital (Heinkel et al. 2001; Chowdhry et al. 2019; Pástor et al. 2021; Avramov et al. 2022; Zerbib 2022; Avramov et al. 2024; Green and Roth 2025; Landier and Lovo 2025; Oehmke and Opp 2025; Dangl et al. 2025; Inderst and Opp 2025; Goldstein et al. 2025). Closest to the portfolio-construction problem, Pedersen et al. (2021) develop the ESG-efficient frontier for investors with ESG preferences, while Lo and Zhang (2024), Lo et al. (2024), and Berg et al. (2024) study portfolio choice with ESG scores and other nonfinancial ranking variables. Impact returns are a natural input to these portfolio-construction models.

The paper is organized as follows. Section 2 develops the supply-and-demand system and derives the contribution multiplier matrix. Section 3 defines net outcome intensities, impact returns, and the Impact Frontier. Sections 4 and 5 describe the data, identification, and estimated slope system. Section 6 constructs the calibrated Impact Frontiers and compares them with simple tilt rules. Section 7 discusses limitations and extensions and Section 8 concludes. Technical derivations and supplemental empirical results are provided in the Online Appendix (henceforth the Appendix).

2 Supply-and-demand system

This section introduces the marginal supply-and-demand system. The Appendix derives it from local market clearing around a benchmark equilibrium with a risk-free asset.

2.1 *One-asset intuition*

It is useful to fix ideas with one asset. The setup is written as a single-period model: Investors supply financial capital at the start of the period based on expected payoffs at the end of the period. An issuer offers claims in exchange for that capital. Market clearing occurs at the start of the period and determines both the price P of those claims and the amount of capital K allocated to the issuer. In this one-asset subsection, K is measured in units of deployable capital, so an increase $\mathrm{d}K$ uses $\mathrm{d}K$ units of investor cash.

Write the issuer's marginal capital demand schedule around a market-clearing point (K^*, P^*) in deviations as

$$\mathrm{d}P = \Phi_D \mathrm{d}K,$$

and the aggregate investor marginal capital supply schedule as

$$\mathrm{d}P = -\Phi_S \mathrm{d}K,$$

where $\mathrm{d}P$ and $\mathrm{d}K$ are local deviations from (P^*, K^*); after a supply-schedule shift is introduced, market clearing determines their equilibrium values. $\Phi_D > 0$ is the issuer-demand slope, and $\Phi_S > 0$ is the aggregate investor-supply slope. A larger positive slope means the quantity of capital is less elastic with respect to price along that schedule.

Figure 1 illustrates the setup. Market clearing implies the equilibrium capital change $\mathrm{d}K = \mathcal{C}^{\mathrm{dir}} \delta k_i$, with direct contribution multiplier

$$\mathcal{C}^{\mathrm{dir}} \equiv \frac{\Phi_S}{\Phi_S + \Phi_D}. \tag{1}$$

The contribution multiplier depends on the relative slopes, not either slope in isolation. The

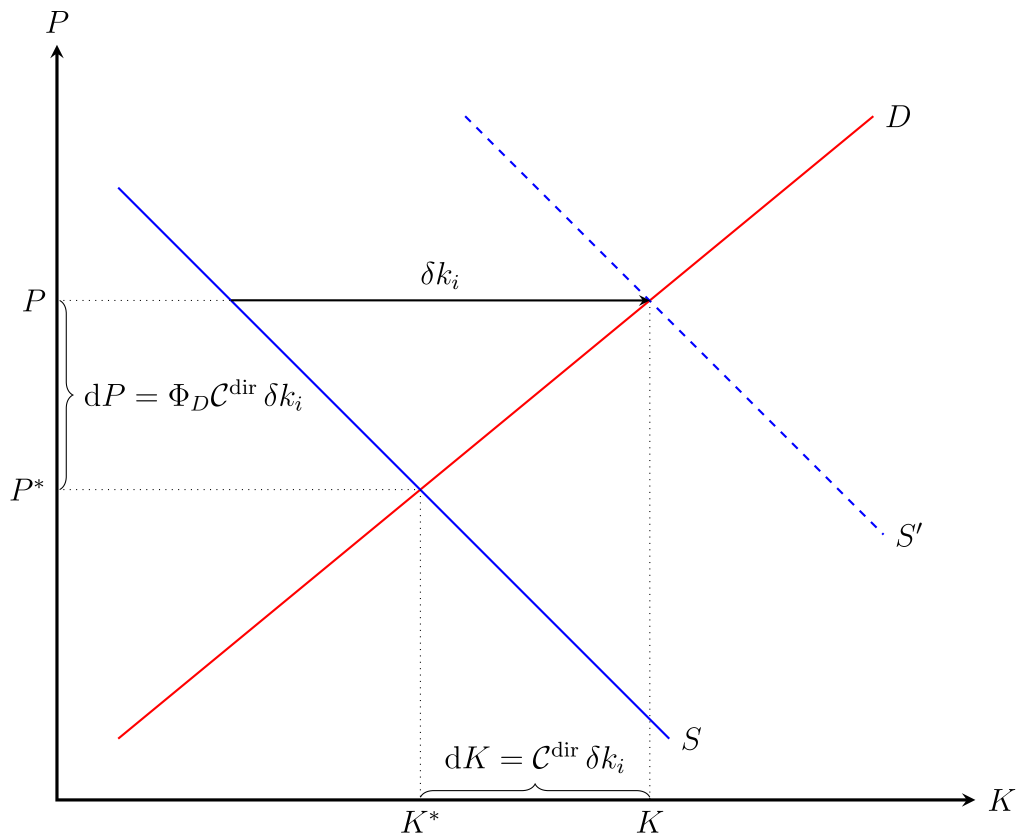

Figure 1: **One-asset intuition for the direct contribution multiplier.** This stylized diagram illustrates the one-asset supply-and-demand system for capital. The issuer-demand schedule D has slope Φ_D, while the investor-supply schedule S has slope $-\Phi_S$ in the (K, P) plane. When investor i shifts their supply schedule by δk_i, with the other investors' schedules and issuer demand held fixed, the aggregate investor supply schedule moves from S to S'. The resulting capital change is $\mathrm{d}K = \mathcal{C}^{\mathrm{dir}} \delta k_i$, with price change $\mathrm{d}P = \Phi_D \mathcal{C}^{\mathrm{dir}} \delta k_i$.

multiplier increases when the investor-supply slope is larger, so other investors absorb less of the tilt, or when the issuer-demand slope is smaller, so more of the tilt becomes issuer capital. The common intuition that capital supplied to smaller or less broadly held firms should have larger effects is an investor-supply statement, and need not hold when there is demand-slope heterogeneity.

Let B denote the aggregate risk-free-asset position of the investors. The equilibrium change is $\mathrm{d}B = -\mathrm{d}K$, holding the risk-free rate fixed. Accounting for the change $\mathrm{d}B$ is important because the risk-free asset may itself support activities, outside the risky-asset market, that produce external outcomes.

The next subsection extends the same logic to multiple assets, where contribution multipliers can differ across assets and a tilt toward one asset can reallocate capital across others.

2.2 Multi-asset supply and demand

Let there be N_A risky assets with price vector $\boldsymbol{P} \in \mathbb{R}^{N_A}$ and aggregate capital vector $\boldsymbol{K} \in \mathbb{R}^{N_A}$ around a benchmark market-clearing point $(\boldsymbol{K}^*, \boldsymbol{P}^*)$. Write the issuer side in deviations as

$$\mathrm{d}\boldsymbol{P} = \boldsymbol{\Phi}_D \, \mathrm{d}\boldsymbol{K}, \tag{2}$$

and the aggregate investor side as

$$\mathrm{d}\boldsymbol{P} = -\boldsymbol{\Phi}_S \, \mathrm{d}\boldsymbol{K}. \tag{3}$$

The matrices $\boldsymbol{\Phi}_S, \boldsymbol{\Phi}_D \in \mathbb{R}^{N_A \times N_A}$ are the supply and demand slope matrices. As in the one-asset case, $\mathrm{d}\boldsymbol{P}$ and $\mathrm{d}\boldsymbol{K}$ denote local deviations whose equilibrium values are solved after the perturbation $\delta \boldsymbol{k}_i$ is introduced.

Consider an investor i with wealth $W_i > 0$ who makes a marginal supply-schedule shift. Suppose they may only shift their schedule on a subset of assets $\mathcal{N}_i \subseteq \{1, \ldots, N_A\}$ called their active opportunity set. Active opportunity sets matter because some investors may be unable or unwilling to adjust positions in every asset. Incomplete information, mandates, benchmark constraints, or other frictions can reduce the marginal investor base for some assets (Merton 1987). Let $\boldsymbol{I}_i \in \mathbb{R}^{|\mathcal{N}_i| \times N_A}$ be the selection matrix for investor i's active opportunity set. The transpose $\boldsymbol{I}_i'$ embeds an active-set shift into the full N_A-asset vector.

A shift $\delta \boldsymbol{k}_i \in \mathbb{R}^{|\mathcal{N}_i|}$ by investor i changes the aggregate investor schedule to

$$\mathrm{d}\boldsymbol{P} = \boldsymbol{\Phi}_S \boldsymbol{I}_i' \, \delta \boldsymbol{k}_i - \boldsymbol{\Phi}_S \, \mathrm{d}\boldsymbol{K}, \tag{4}$$

with the other investors' supply curves held fixed.

Proposition 1 (Contribution multiplier matrix)**.** Assume the total slope matrix $\boldsymbol{\Phi}_T \equiv \boldsymbol{\Phi}_S + \boldsymbol{\Phi}_D$ is nonsingular. Under (2) and (4), the generated issuer capital change is

$$\mathrm{d}\boldsymbol{K} = \boldsymbol{\mathcal{C}} \boldsymbol{I}_i' \, \delta \boldsymbol{k}_i, \tag{5}$$

where the contribution multiplier matrix is

$$\boldsymbol{\mathcal{C}} \equiv \boldsymbol{\Phi}_T^{-1} \boldsymbol{\Phi}_S. \tag{6}$$

Proof. Equating the issuer response $\mathrm{d}\boldsymbol{P} = \boldsymbol{\Phi}_D \mathrm{d}\boldsymbol{K}$ with the shifted supply response $\mathrm{d}\boldsymbol{P} = \boldsymbol{\Phi}_S \boldsymbol{I}_i' \delta \boldsymbol{k}_i - \boldsymbol{\Phi}_S \mathrm{d}\boldsymbol{K}$ implies $\boldsymbol{\Phi}_T \mathrm{d}\boldsymbol{K} = \boldsymbol{\Phi}_S \boldsymbol{I}_i' \delta \boldsymbol{k}_i$, hence $\mathrm{d}\boldsymbol{K} = \boldsymbol{\Phi}_T^{-1} \boldsymbol{\Phi}_S \boldsymbol{I}_i' \delta \boldsymbol{k}_i = \boldsymbol{\mathcal{C}} \boldsymbol{I}_i' \delta \boldsymbol{k}_i$. $\square$

Equation (6) is the multi-asset analogue of (1). Column n reports the equilibrium issuer-capital response to an investor's supply-schedule shift toward asset n. The diagonal entries report the full own-asset equilibrium response, and the off-diagonal entries report the capital changes associated with other assets induced by the same tilt.

The schedules above assume the risk-free-asset supply is highly elastic and the risk-free rate is fixed. Under this baseline the risk-free-asset accounting is

$$\mathrm{d}B = -\mathbf{1}' \mathrm{d}\boldsymbol{K}. \tag{7}$$

The Appendix allows a general cash requirement $\boldsymbol{\Pi}' \mathrm{d}\boldsymbol{K}$, with the baseline corresponding to $\boldsymbol{\Pi} = \mathbf{1}$, and includes the inelastic case in which the risk-free rate adjusts.

2.3 Direct contribution multipliers

The direct contribution multiplier for asset n is

$$\mathcal{C}_n^{\text{dir}} \equiv \frac{\Phi_{S,nn}}{\Phi_{T,nn}}.$$

It is the one-asset contribution multiplier implied by the diagonal supply and demand slopes. Let the vector $\boldsymbol{\mathcal{C}}^{\text{dir}}$ collect these multipliers. The difference $\boldsymbol{\mathcal{C}} - \text{diag}(\boldsymbol{\mathcal{C}}^{\text{dir}})$ captures the system response: off-diagonal capital reallocations and any difference between the full own-asset response $\mathcal{C}_{nn}$ and the direct contribution multiplier $\mathcal{C}_n^{\text{dir}}$.

2.4 Aggregate investor-supply slope matrix

The Appendix derives the exact aggregation of individual investor-supply slopes and a one-factor approximation when investors have different active opportunity sets. In that approximation,

$$\boldsymbol{\Phi}_S \approx \underbrace{\bar{\gamma}\boldsymbol{\Sigma}_{\text{val}}}_{\boldsymbol{\Phi}_S^M} + \underbrace{\bar{\gamma}\,\text{diag}\left(\boldsymbol{q}^{-1} - \mathbf{1}\right)\text{diag}\left(1 - \boldsymbol{\rho}^2\right)\text{diag}(\boldsymbol{\sigma}_{\text{val}}^2)}_{\boldsymbol{\Phi}_S^N}, \tag{8}$$

where $\bar{\gamma}$ is aggregate marginal risk aversion in dollar-holding units, $\boldsymbol{\Sigma}_{\text{val}}$ is the covariance matrix of asset values at the horizon, with diagonal $\boldsymbol{\sigma}_{\text{val}}^2$, $\boldsymbol{\rho}$ contains correlations with the common payoff factor, represented by the market portfolio in the empirical specification, and $q_n \in (0, 1]$ is the active-investor share for asset n, weighted by inverse marginal risk aversion. $\boldsymbol{\Phi}_S^M$ is the market-risk matrix, and $\boldsymbol{\Phi}_S^N$ is the diagonal non-market-risk matrix. A lower q_n raises the investor-supply slope associated with issuer n's residual risk. This one-factor approximation motivates the non-market-risk term used in the empirical supply equation. The expression is exact under common active sets or when all investors can trade a forward contract on the common factor, as in Merton (1987).

2.5 Sharpe-ratio representation

To express the local supply schedule in Sharpe-ratio units, let $\mu_n = \mathbb{E}[r_n] - r_f$ and $\lambda_n = \mu_n/\sigma_{R,n}$, where $\sigma_{R,n}$ is return volatility, and let $\rho_{M,n}$ denote asset n's return correlation with the market portfolio. Define issuer risk capital and market risk capital by $Q_n \equiv \sigma_{R,n} V_n$ and $Q_M \equiv \sigma_M V_M$. The risk-capital share is $\rho^{\text{int}}_{M,n} \equiv Q_n/Q_M$. This scaling turns the market-risk block into market correlation and the residual supply-slope block into risk-capital share.

Inserting the one-factor aggregate supply slope matrix from (8) into the supply equation and expressing it in λ-ρ terms yields

$$\mathrm{d}\lambda_n = \mathrm{d}\left[\underbrace{\bar{\gamma} Q_M \rho_{M,n}}_{\text{Market risk}} + \underbrace{\bar{\gamma} Q_M (q_n^{-1} - 1)(1 - \rho^2_{M,n})\rho^{\text{int}}_{M,n}}_{\text{Non-market risk}}\right]. \tag{9}$$

This separates the market-risk part of the local investor-supply schedule from the non-market-risk part implied by the difference between $\boldsymbol{\Phi}_S$ and $\bar{\gamma}\boldsymbol{\Sigma}_{\text{val}}$.

3 Outcomes and the Impact Frontier

This section defines net outcome intensities, impact returns, and the Impact Frontier.

3.1 Net outcome intensities

Let X be an external outcome variable, such as cumulative greenhouse-gas emissions avoided, a biodiversity outcome, a health outcome, or another real outcome. The timing of X is application-specific and is left abstract here. Issuer n's outcome intensity is defined as the marginal effect of issuer capital on that outcome,

$$g_n \equiv \frac{\partial X}{\partial K_n}. \tag{10}$$

A portfolio tilt changes issuer capital and risk-free-asset holdings. Let $\bar{g}_B$ denote the outcome intensity assigned to the risk-free asset. Combining (7) with (10), any issuer-capital change $\mathrm{d}\boldsymbol{K}$ changes the outcome by

$$\mathrm{d}X = \boldsymbol{g}'\mathrm{d}\boldsymbol{K} + \bar{g}_B \mathrm{d}B = (\boldsymbol{g}_{\text{net}})'\mathrm{d}\boldsymbol{K}, \tag{11}$$

where

$$\boldsymbol{g}_{\text{net}} \equiv \boldsymbol{g} - \bar{g}_B \mathbf{1} \tag{12}$$

is the vector of net outcome intensities. The netting makes the calculation invariant to adding a constant to every gross outcome intensity. In the empirical baseline, $\bar{g}_B$ is the total-assets-weighted mean outcome intensity, so a shift proportional to the observed issuer-capital allocation has zero net outcome change. This makes the frontier reflect outcome change from reallocating capital relative to the observed allocation, rather than from the potentially arbitrary level of the gross outcome-intensity scale. The Appendix shows how alternative risk-free-asset supply assumptions can be captured through the choice of $\bar{g}_B$.

3.2 Impact returns and portfolio impact returns

For the investor i introduced above, a supply-schedule shift $\delta\boldsymbol{k}_i$ generates, combining (5) and (12), the expected outcome change $\Delta X_i = \boldsymbol{h}'\boldsymbol{I}_i'\delta\boldsymbol{k}_i$, depending on the impact intensity vector

$$\boldsymbol{h} \equiv \boldsymbol{C}'\boldsymbol{g}_{\text{net}}, \tag{13}$$

which represents contribution-adjusted outcome intensities.

Impact returns are then defined as

$$\boldsymbol{r}^I \equiv \text{diag}(\boldsymbol{P}^*)^{-1}\boldsymbol{h}, \tag{14}$$

and represent expected outcome change per dollar of position value in each asset after applying the contribution multiplier matrix to the net outcome intensities. These are returns only in the sense of outcome units per dollar of position value; they are not expected financial returns.

For investor i, define the active dollar tilt $\mathrm{d}\boldsymbol{V}_i \equiv \mathrm{diag}(\boldsymbol{I}_i\boldsymbol{P}^*)\delta\boldsymbol{k}_i$, and active portfolio-weight tilt

$$\mathrm{d}\boldsymbol{w}_i \equiv \mathrm{d}\boldsymbol{V}_i/W_i. \tag{15}$$

Then the expected outcome change is $\Delta X_i(\mathrm{d}\boldsymbol{V}_i) = W_i\boldsymbol{r}^{I\prime}\boldsymbol{I}_i'\mathrm{d}\boldsymbol{w}_i$. The investor's portfolio impact return,

$$r^I_{p,i} \equiv \boldsymbol{r}^{I\prime}\boldsymbol{I}_i'\mathrm{d}\boldsymbol{w}_i \tag{16}$$

is the expected outcome change per unit investor wealth.

3.3 Impact Frontier

The frontier evaluates marginal active tilts around investor i's default portfolio, before assigning a shadow value to outcome X. At that portfolio, feasible active tilts have no linear CE cost, so the leading CE cost is quadratic. Let $\boldsymbol{\Phi}_i^V$ denote investor i's active supply slope matrix in dollar units, with dimensions equal to the investor's active opportunity set. It describes the local expected return investor i requires to supply an additional active dollar around that default portfolio. Since the frontier is reported in CE-return units, define the effective covariance matrix

$$\boldsymbol{\Sigma}_i^{\mathrm{eff}} \equiv W_i\boldsymbol{\Phi}_i^V. \tag{17}$$

This effective covariance matrix defines the CE-return cost of a tilt. In standard mean-variance analysis, it reduces to the return covariance matrix scaled by relative risk aversion. In general, it may also include terms that represent other costs and frictions.

For the empirical calculations, the paper uses two CE specifications. Market-Risk CE applies equation (17) to the market-risk part of the aggregate supply slope, while Aggregate-Supply CE applies it to the full aggregate supply slope, using aggregate market value V_M as the wealth scalar. When individual supply slopes are proportional after wealth scaling, Aggregate-Supply CE is the corresponding average effective covariance matrix implicit in the aggregate marginal supply schedule. It will not describe every investor individually, but it is a natural reference when the market portfolio is the investor's default portfolio.

For an active portfolio-weight tilt $\mathrm{d}\boldsymbol{w}_i$, define the local CE-return cost as

$$\Delta r_i^{CE}(\mathrm{d}\boldsymbol{w}_i) \equiv \frac{1}{2}\mathrm{d}\boldsymbol{w}_i'\boldsymbol{\Sigma}_i^{\text{eff}}\mathrm{d}\boldsymbol{w}_i. \tag{18}$$

The corresponding relative CE return is $-\Delta r_i^{CE}$.

Proposition 2 (Impact Frontier)**.** Let $\eta_i^2 \equiv \boldsymbol{r}^{I\prime}\boldsymbol{I}_i'(\boldsymbol{\Sigma}_i^{\text{eff}})^{-1}\boldsymbol{I}_i\boldsymbol{r}^I$. Each point on the Impact Frontier corresponds to an active portfolio-weight tilt

$$\mathrm{d}\boldsymbol{w}_i = \gamma_i^X(\boldsymbol{\Sigma}_i^{\text{eff}})^{-1}\boldsymbol{I}_i\boldsymbol{r}^I.$$

For target portfolio impact return $r_{p,i}^I$,

$$\gamma_i^X = \frac{r_{p,i}^I}{\eta_i^2}, \qquad \Delta r_i^{CE} = \frac{1}{2}\frac{(r_{p,i}^I)^2}{\eta_i^2}, \tag{19}$$

or for CE-return cost Δr_i^{CE},

$$\gamma_i^X = \sqrt{\frac{2\Delta r_i^{CE}}{\eta_i^2}}, \qquad r_{p,i}^I = \sqrt{2\Delta r_i^{CE}}\,\eta_i. \tag{20}$$

The scalar η_i^2 measures frontier strength. A higher value implies more portfolio impact return at a fixed CE-return cost, or equivalently a lower CE-return cost for a target portfolio impact return.

3.4 Below the frontier

A portfolio is impact-inefficient if it lies below the Impact Frontier: another feasible portfolio produces more portfolio impact return for the same CE-return cost, or the same portfolio impact return at lower CE-return cost.

The paper uses two alternative tilts to isolate how direct contribution multipliers and the remaining system response affect recovery. Each tilt can be represented by replacing the full contribution multiplier matrix with an alternative $\widetilde{\boldsymbol{\mathcal{C}}}_{\text{mis}}$. Relative to the Impact Frontier, these rules use misspecified impact returns

$$\tilde{\boldsymbol{r}}^I_{\text{mis}} \equiv \operatorname{diag}(\boldsymbol{P}^*)^{-1}\widetilde{\boldsymbol{\mathcal{C}}}'_{\text{mis}}\boldsymbol{g}_{\text{net}}. \tag{21}$$

The net-outcome and direct cases are examples of this choice: the net-outcome tilt sets $\widetilde{\boldsymbol{\mathcal{C}}}_{\text{mis}} = \boldsymbol{I}_{N_A}$, so its implied impact returns use $\boldsymbol{g}_{\text{net}}$ directly. The direct tilt sets $\widetilde{\boldsymbol{\mathcal{C}}}_{\text{mis}} = \operatorname{diag}(\boldsymbol{\mathcal{C}}^{\text{dir}})$, using the direct contribution multipliers $\boldsymbol{\mathcal{C}}^{\text{dir}}$ but omitting system response.

At CE-return cost $\Delta r_i^{CE} > 0$, the expected portfolio impact return of the comparison tilt is $r^I_{p,i} = \sqrt{2\Delta r_i^{CE}}\,\tilde{\eta}_{i,\text{mis}}$, where

$$\tilde{\eta}_{i,\text{mis}} \equiv \frac{(\boldsymbol{r}^I)'\boldsymbol{I}_i'(\boldsymbol{\Sigma}_i^{\text{eff}})^{-1}\boldsymbol{I}_i\tilde{\boldsymbol{r}}^I_{\text{mis}}}{\left[(\tilde{\boldsymbol{r}}^I_{\text{mis}})'\boldsymbol{I}_i'(\boldsymbol{\Sigma}_i^{\text{eff}})^{-1}\boldsymbol{I}_i\tilde{\boldsymbol{r}}^I_{\text{mis}}\right]^{1/2}}. \tag{22}$$

The recovery ratio for equal-CE between the inefficient tilt and the frontier is

$$\tilde{\rho}_i \equiv \frac{\tilde{\eta}_{i,\text{mis}}}{\eta_i}. \tag{23}$$

A value of one means the inefficient tilt reproduces the Impact Frontier, zero means it produces no outcome change, and a negative value means it pushes in the wrong direction. The Appendix uses eigenanalysis to derive tractable formulas and approximations for worst-case recoveries reported in Section 6.

When $\tilde{\eta}_{i,\text{mis}} > 0$, this also allows an equal-impact cost comparison. For a target portfolio impact return $r^I_{p,i} > 0$, the CE-return-cost multiple required to match the frontier result is

$$\frac{\Delta r^{CE}_{i,\text{mis}}}{\Delta r^{CE}_{i,\text{IF}}} = \frac{\eta_i^2}{\tilde{\eta}^2_{i,\text{mis}}} = \frac{1}{\tilde{\rho}_i^2}. \tag{24}$$

Thus a tilt that recovers one half of the frontier portfolio impact return at the same CE-return cost requires four times the frontier CE-return cost to reach the same portfolio impact return. If $\tilde{\eta}_{i,\text{mis}} \leq 0$, positive scaling of the tilt cannot produce a positive portfolio impact return.

3.5 Frontier-strength decomposition

Frontier strength and recovery ratios are quadratic forms in the contribution multiplier matrix, the effective covariance matrix, and the outcome-intensity vector. The Appendix diagonalizes these quadratic forms. In that diagonalization, a mode is an eigenvector of the frontier-strength quadratic form after the active set and effective covariance matrix are fixed. It is a direction in outcome-intensity space, not a firm, portfolio, or asset-pricing factor.

The resulting log-frontier-strength decomposition is

$$\log \eta^2 = 2S_g + 2G_{\text{dist}} + 2C + B - R + \epsilon. \tag{25}$$

The S_g term captures the variance of the net outcome intensity. The G_{dist} term captures whether the normalized outcome-intensity vector loads on high-frontier-strength directions. The C term captures the average contribution strength along those directions: how strongly the contribution multiplier matrix converts supply shifts into issuer-capital changes. The B term is the entropy of the modal contribution shares, so it is larger when frontier strength is spread across several directions rather than concentrated in one. The R term captures the CE-return cost of moving capital along those directions, and ϵ records the modal-factorization residual. Section 6 uses this decomposition to interpret the empirical frontier-strength differences across outcome directions.

4 Data and empirical estimation

This section describes the empirical supply-and-demand system that provides the estimated slopes which enter the contribution multiplier matrix and the frontier calculations.

4.1 Econometric specification

Variables are indexed by issuer n and panel date t. The vectors $\boldsymbol{z}_{S,n,t}$ and $\boldsymbol{z}_{D,n,t}$ collect the included investor-side and issuer-side variables, and $\hat{\lambda}_{n,t}$ is the forward Sharpe ratio. The investor-side risk terms use the issuer's correlation with the market portfolio, $\hat{\rho}_{M,n,t}$, and the risk-capital share $\hat{\rho}^{\text{int}}_{M,n,t}$ defined in Section 2.5. Both equations include year and industry fixed effects. The investor-side equation is a level regression that estimates the corresponding local supply slopes using cross-sectional variation in the forward Sharpe ratios. The coefficients Δ_M and Δ_N correspond to the market-risk and non-market-risk terms in (9). The included variables and fixed effects absorb additive Sharpe-ratio variation not assigned to those two slope terms. The investor-side equation is

$$\hat{\lambda}_{n,t} = \boldsymbol{z}'_{S,n,t}\beta_S + \Delta_M \hat{\rho}_{M,n,t} + \Delta_N \hat{z}^N_{n,t} + u_{S,n,t}, \tag{26}$$

where the non-market-risk term with coefficient Δ_N is

$$\hat{z}^N_{n,t} = (1 - p^{TA}_{n,t})(1 - \hat{\rho}^2_{M,n,t})\hat{\rho}^{\text{int}}_{M,n,t}. \tag{27}$$

The issuer-side equation is

$$\hat{\lambda}_{n,t} = \boldsymbol{z}'_{D,n,t}\beta_D + \Delta_D \log(TA_{n,t}/TA_{n,t-5}) + u_{D,n,t}. \tag{28}$$

The system estimates local multi-year supply and demand slopes around the observed market-clearing point. At panel date t, the investor-supply equation uses forward return variables, while the issuer-demand equation uses total-assets growth ending at the panel date.

Empirically, let $Q_{n,t} \equiv \hat{\sigma}_{R,n,t}V_{n,t}$ denote issuer n's risk capital and let $Q_{M,t} \equiv \hat{\sigma}_{M,t}V_{M,t}$ denote market risk capital. The risk-capital share is

$$\hat{\rho}^{\text{int}}_{M,n,t} \equiv Q_{n,t}/Q_{M,t}.$$

The term $(1-\hat{\rho}^2_{M,n,t})$ is the residual-risk share after projecting issuer n's return on the market return. Because q_n is unobserved, the empirical non-market-risk term uses $1 - p^{TA}_{n,t}$ as a neglectedness proxy for thinner marginal active participation, where $p^{TA}_{n,t}$ is the percentile of horizon-lagged total assets, with $p^{TA}_{n,t} = 1$ for the largest issuers and $p^{TA}_{n,t} = 0$ for the smallest. The proxy is not an estimate of $q_n^{-1} - 1$: even when large investors can trade all sample issuers, costly research and lower mandate relevance may make smaller issuers less likely to attract active attention. The specification tests whether non-market risk is priced more strongly where this proxy is higher, not whether idiosyncratic volatility earns a generic premium.

Total assets is the baseline proxy for issuer capital. Thus, Δ_D is the fitted Sharpe-ratio change associated with log capital growth. Holding $TA_{n,t-5}$ fixed, the log specification implies

$$\left.\frac{\partial \hat{\lambda}_{n,t}}{\partial TA}\right|_{TA=TA_{n,t}} = \Delta_D \left.\frac{\partial \log(TA/TA_{n,t-5})}{\partial TA}\right|_{TA=TA_{n,t}} = \frac{\Delta_D}{TA_{n,t}}. \tag{29}$$

This makes the demand coefficient comparable across large and small issuers, while a level-change specification would instead impose a common dollar-capital slope. The timing makes the implied level slope $\Delta_D/TA_{n,t}$ a local cross-sectional calibration around t. The Appendix tests alternative specifications and timing.

The forward Sharpe ratio, market correlation, volatility, and non-market-risk term are generated from finite return windows; reported system standard errors condition on these generated quantities.

4.2 Data

Sample. The sample is built from the merged CRSP-Compustat universe. When multiple securities are available, the construction keeps the main listed equity for each issuer. The baseline return frequency is monthly. The baseline pools the December 31 cross-sections for 2004, 2009, 2014, and 2019. Return-based variables use five-year forward windows; the baseline issuer-demand quantity is five-year lagged total-assets growth. The longer return window reduces short-horizon noise in Sharpe-ratio, correlation, and volatility estimates.

The final estimation sample requiring the Environmental score contains 4120 issuer-year observations. It covers 1910 distinct issuers. The matched no-Environmental specification in the final 3SLS column uses the same issuer-years by construction, while a separate expanded no-score diagnostic contains 7051 issuer-year observations because it does not impose a score-availability screen. The 2019 cross-section used for the baseline frontier analyses contains 1,200 issuers after requiring complete monthly returns over the five-year covariance window.

These screens make the calibration conditional on sufficient accounting, score, and return data around the panel date.

Outcome-score proxies. The frontier construction applies to any outcome-intensity vector. The baseline applications use the MSCI Social, Environmental, Governance, and Carbon Emissions scores as illustrative outcome-intensity proxies with broad coverage in this public-equity universe. These scores are designed primarily for investment analysis rather than marginal external-outcome measurement; here they illustrate how the contribution multiplier matrix transforms a chosen outcome-intensity vector. When a frontier cross-section includes an issuer without a particular raw score, the implementation assigns that issuer the score's total-assets-weighted mean before risk-free-asset netting. Equivalently, missing score entries receive zero net outcome intensity for that score direction.

Table 1 reports summary statistics for the full panel and the 2019 frontier cross-section. The Appendix reports sample filters, variable construction, OLS and 2SLS estimates, first-stage tests, and demand-side specification tests.

4.3 Identification

The baseline system uses equation-specific endogenous variables, included controls, and excluded instruments. The restrictions are imposed on the jointly estimated supply-and-demand system, not on either equation alone.

Investor-supply identification. The investor-supply equation includes two endogenous variables: market risk and the constructed non-market-risk term in (27). It controls for the Environmental score as an incidental estimation variable, business-equity correlation, and year and industry fixed effects. Lagged Sharpe ratio, lagged total-assets percentile, and profitability are excluded instruments for the investor-supply equation. The restriction is that, conditional on the included controls and fixed effects, these variables shift the endogenous supply-side risk terms but do not directly add an independent required-return component. Because $1 - p_{n,t}^{TA}$ is a predetermined component of the constructed non-market-risk term, the lagged total-assets percentile is allowed to affect required returns through that endogenous risk term, but not as an independent required-return term. For lagged Sharpe ratio, the exclusion restriction is that its predictive content for forward Sharpe ratios operates through

	Full Panel		2019 Cross-Section	
	Mean	SD	Mean	SD
Market portfolio				
Risk premium (% per year)	8.97		11.5	
Volatility (% per year)	15.3		18.9	
Sharpe ratio	0.624		0.611	
Assets and issuers				
Risk premium (% per year)	6.1	18.4	2.46	19
Volatility (% per year)	40.2	23.1	48.4	32.6
Sharpe ratio (forward 5Y)	0.249	0.48	0.104	0.383
Market correlation	0.5	0.17	0.515	0.17
Market cap ($ billion)	12.8	48.2	18.6	73.9
Book equity ($ billion)	3.81	12.8	4.4	15.5
Total assets ($ billion)	9.25	32.9	11.4	35.8
Issuer-demand variable				
Log total-assets growth, $t-5$ to t	0.54	0.751	0.544	0.854
Supply-side and identification variables				
Business equity correlation	−0.0524	0.204	−0.106	0.184
Profitability	0.282	2.46	0.285	1.96
Cash flow volatility	3.24	21.1	3.63	32.8
Lagged Sharpe ratio	0.267	0.512	0.289	0.55

Table 1: **Summary Statistics.** This table reports equal-weighted summary statistics for the full panel and the 2019 cross-section. The full panel pools the December 31 cross-sections for 2004, 2009, 2014, and 2019 after applying the baseline estimation screens requiring the Environmental score; it contains 4120 issuer-year observations and 1910 distinct issuers. The 2019 cross-section is used in the main frontier figures and contains 1,200 issuers. Market-portfolio statistics are computed over the forward five-year windows aligned with the panel dates. Asset return, volatility, Sharpe-ratio, and market-correlation variables use forward returns; issuer accounting variables are measured at the panel date or over the stated lag window.

the endogenous risk terms after the included controls and fixed effects are held fixed; the anomaly and pre-panel timing diagnostics in the Appendix stress-test this restriction.

Issuer-demand identification. The issuer-demand equation treats log total-assets growth as endogenous and controls for lagged Sharpe ratio and the same fixed effects. The Environmental score, business-equity correlation, and lagged total-assets percentile are excluded instruments for issuer demand. For the Environmental score and lagged total-assets percentile, the restriction is that, conditional on lagged Sharpe ratio and fixed effects, they move required returns through investor supply rather than directly shifting issuer demand. Business-equity correlation is a forward-horizon risk variable in the baseline; the pre-t timing diagnostic repeats the system with a pre-panel analogue to test whether the result relies on forward return-window overlap. The coefficient calibrates the local cross-sectional issuer-demand slope entering the contribution multiplier matrix. The Appendix reports alternative demand quantities and the resulting frontier calculations.

5 Estimated supply and demand system

This section estimates the baseline multi-year supply-and-demand system used in the frontier analysis. The baseline uses the 2004, 2009, 2014, and 2019 panel years, 60-month windows, correlations of monthly returns, and total-assets growth as the issuer-demand quantity.

5.1 Estimated slopes and elasticity magnitudes

Table 2 reports three-stage least squares estimates of the supply-and-demand system (Zellner and Theil 1962). 3SLS is the baseline method because the equation residuals may be correlated. The Appendix reports equation-by-equation 2SLS estimates. Their close agreement indicates that the baseline slopes are driven by the IV correction rather than the additional cross-equation weighting in 3SLS.

The non-baseline columns are sensitivity checks rather than alternative frontier inputs. Removing cash-flow volatility, removing profitability from the investor-supply excluded-instrument list, and removing the Environmental score on the matched sample leave the market-risk and non-market-risk coefficients positive and strongly estimated. The issuer-

	Dependent variable: Sharpe ratio			
	(1)	(2)	(3)	(4)
Investor Supply				
Market correlation (ρ_M)	1.35***	1.35***	1.34***	1.35***
	(0.11)	(0.11)	(0.11)	(0.11)
Non-market risk term	726***	725***	789***	713***
	(190)	(190)	(190)	(190)
Environmental score	-0.0053**	-0.0053**	-0.0055**	
	(0.0024)	(0.0024)	(0.0024)	
Cash-flow volatility	0.009			0.009
	(0.021)			(0.022)
Business-equity correlation	0.191***	0.191***	0.194***	0.191***
	(0.028)	(0.028)	(0.028)	(0.028)
Issuer Demand				
Log total-assets growth, $t-5$ to t	-0.235***	-0.235***	-0.235***	-0.241***
	(0.021)	(0.021)	(0.021)	(0.021)
Sharpe ratio (lagged)	0.156***	0.156***	0.165***	0.157***
	(0.016)	(0.016)	(0.015)	(0.016)
Excluded instruments				
Investor supply				
Lagged Sharpe ratio	Yes	Yes	Yes	Yes
Lagged total-assets percentile	Yes	Yes	Yes	Yes
Profitability	Yes	Yes	No	Yes
Issuer demand				
Environmental score	Yes	Yes	Yes	No
Business-equity correlation	Yes	Yes	Yes	Yes
Lagged total-assets percentile	Yes	Yes	Yes	Yes
Diagnostics				
Sample				
Number of observations	4120	4120	4315	4120
Distinct issuers	1910	1910	1974	1910
Investor supply				
First-stage F (min)	77	76	120	76
Wu-Hausman	56	57	61	58
Sargan p-value	0.36	0.36	–	0.36
Issuer demand				
First-stage F	370	370	360	550
Wu-Hausman	97	97	100	100
Sargan p-value	0.10	0.10	0.14	0.59

Table 2: **3SLS estimates of the supply and demand system.** This table reports three-stage least squares estimates of the simultaneous supply-and-demand system, with forward Sharpe ratio on the left-hand side of both equations. Column 1 is the baseline specification used to construct the main frontier calculations. Column 2 removes cash-flow volatility. Column 3 removes profitability from the investor-supply excluded-instrument list while keeping cash-flow volatility out. Column 4 removes the Environmental score from the investor-side equation and from the issuer-demand excluded-instrument list on the matched Column 1 sample. All specifications include year and industry fixed effects. The bottom blocks report excluded instruments and diagnostics; included controls are reported in the coefficient rows. For investor supply, the first-stage statistic reports the minimum across the two endogenous risk terms. Sargan rows report p-values where overidentification is available; – denotes exact identification or an unavailable test. Parentheses report unclustered 3SLS system standard errors. Stars denote statistical significance at the 0.10, 0.05, and 0.01 levels.

demand coefficient remains negative in all four columns. The frontier calculations use column 1 as the baseline estimation-variable specification; this choice does not give the Environmental score priority as an outcome direction in the frontier applications below.

The positive market-risk coefficient means required Sharpe ratios rise with market exposure. The positive non-market-risk coefficient means required Sharpe ratios also rise with residual risk interacted with the neglectedness proxy. The negative issuer-demand coefficient means higher required Sharpe ratios are associated with lower capital growth, consistent with the investment-CAPM prediction that, all else equal, higher investment is associated with lower expected returns (Zhang 2017). After the transformations presented in the Appendix, these coefficients determine the investor-supply and issuer-demand slope matrices used to construct the contribution multiplier matrix.

The implied supply and demand slopes are economically plausible and sit within a middle range relative to estimates in the broader literature for both sides. For the market portfolio, the baseline supply slope is 0.50, so the corresponding price-elasticity magnitude is 2.0. Removing the non-market-risk term raises the corresponding market-portfolio supply elasticity to 3.7. On the issuer side, the median diagonal demand elasticity is 7.3, and the implied market-portfolio demand elasticity is 7.4. Figure 2 places these calibrated slopes alongside selected literature estimates and the Appendix reports further details. All points are elasticities on either the investor-supply or issuer-demand side, but they are not estimates of the same parameter: they differ in horizon, aggregation level, identifying variation, and structural interpretation.

The Appendix reports sensitivity tests for the slope estimates. Issuer-clustered HC1 standard errors leave the three baseline slope coefficients strongly estimated, and overidentification tests do not reject the baseline instrument restrictions. The first stages are most sensitive to removing lagged Sharpe ratio or lagged total-assets percentile. The issuer-demand coefficient remains negative across the reported demand quantities and timing variants, but the implied demand slope and contribution multipliers change in scale. On the supply side, the non-market-risk coefficient remains positive under anomaly, size, factor-beta, profitability, investment, and trading-coverage tests. It becomes smaller and imprecise when an unrestricted additive neglectedness-proxy term absorbs the same size-related variation, and it

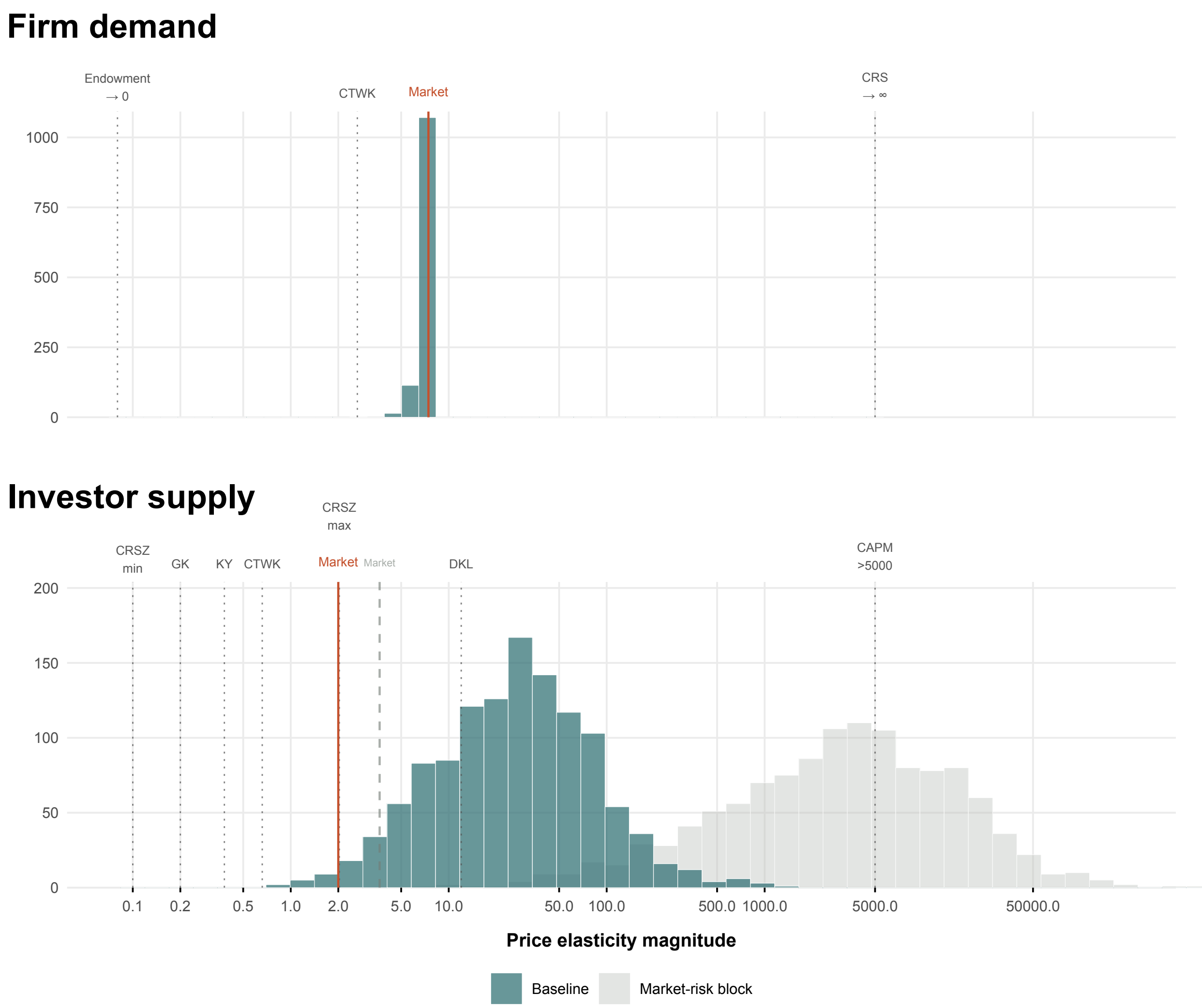

Figure 2: **Elasticity magnitudes in the baseline calibration and selected literature comparisons.** The figure presents price-elasticity magnitudes on a log scale. Literature markers are scale comparisons, not estimates of a common parameter; they differ by horizon, aggregation level, source of variation, and structural interpretation. In the demand panel, the endowment-economy marker denotes the zero issuer-demand-elasticity limit, while the CRS marker denotes the infinite issuer-demand-elasticity limit. Because this paper uses a capital-supply orientation, investor asset-demand estimates are plotted as investor-side supply comparisons. The histograms report firm-level elasticities implied by the baseline calibrated supply-and-demand matrices, while the Market lines indicate the corresponding aggregate market-portfolio elasticities. The baseline results are in teal and red; the light-gray histogram and dashed market line report the market-risk-block decomposition, which holds the baseline coefficients fixed and sets Φ_S^N to zero without re-estimation. The vertical scale counts firms in each histogram bin. Labels abbreviate Gabaix and Koijen (2021) (GK), Koijen and Yogo (2019) (KY), Cassella et al. (2026) (CRSZ), Choi et al. (2025) (CTWK), Davis et al. (2025) (DKL), and constant returns to scale (CRS).

weakens at daily frequency. Pre-t risk variables preserve the coefficient signs. Horizon tests show that shorter-horizon impact-intensity vectors are highly correlated with the five-year vector. Issuer demand becomes more elastic at longer horizons, and the non-market-risk coefficient generally falls, consistent with issuer capital adjusting over longer windows and non-market-risk frictions weakening over longer windows. The market-risk coefficient moves differently, so the estimated slope systems do not rescale by a common factor and the resulting contribution multipliers are horizon-specific.

5.2 *Contribution multiplier heterogeneity*

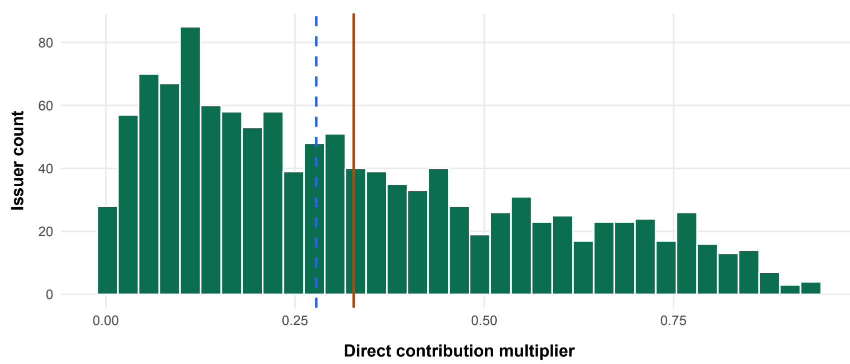

Figure 3: **Direct contribution multipliers.** The figure reports the cross-sectional distribution of direct contribution multipliers $\mathcal{C}_n^{\text{dir}} = \Phi_{S,nn}/\Phi_{T,nn}$ for the baseline frontier cross-section. The solid orange and dashed blue vertical lines mark the equal-weight mean and median.

The estimated slopes imply substantial cross-issuer variation in the direct contribution multipliers. Figure 3 reports these multipliers for the baseline frontier cross-section. The distribution shows dispersion throughout the range from zero to one, with an equal-weight median of 0.28. The value-weighted mean is 0.65. The difference between the equal-weight and value-weighted summaries reflects higher estimated direct contribution multipliers among larger issuers and is not concentrated in a single issuer. Because contribution multipliers are ratios of slopes, their distribution reflects both sides of the market: the non-market-risk term raises investor-supply slopes for smaller, riskier issuers, while the log-growth demand specification implies smaller level demand slopes and higher multipliers for larger issuers.

6 Frontier results

This section combines the estimated supply and demand slope matrices, net outcome intensities, and the effective covariance matrix to construct impact returns and the Impact Frontier. It then compares the frontier with direct and net-outcome tilts. In the public-equity calibration, the investor-side coefficients determine $\mathbf{\Phi}_S$ and the issuer-side coefficient determines $\mathbf{\Phi}_D$; together they determine the contribution multiplier matrix used to construct the frontiers. The Appendix details the market-value scaling, risk-free-asset netting, and matrix construction.

6.1 Frontier scale across outcome directions

The baseline frontier is computed from the column 1 slope estimates in Table 2 under Aggregate-Supply CE. Figure 4 compares frontier scale across the four outcome categories and two normalized illustrative distributions. On their natural score scales, Environmental and Carbon Emissions have stronger frontiers than Social and Governance. Panel B normalizes each net outcome intensity to unit total-assets-weighted standard deviation and shows that the observed score proxies occupy only part of the range permitted by the calibrated system; the constructed distributions are illustrative extremes, not alternative score estimates.

For the score-based intensities, portfolio impact return is the score-weighted issuer-capital change per unit investor wealth. A value of 0.1 means that a tilt by an investor with wealth W_i changes score-weighted issuer capital by $0.1W_i$. As an example of what this means in more intuitive units, the Carbon Emissions table in Section 7 converts the portfolio impact return into illustrative physical units.

Table 3 decomposes the differences in frontier scale into outcome-intensity scale, outcome-distribution alignment, contribution strength, modal breadth, CE-return cost, and the modal-factorization residual. The observed frontiers draw on many modes rather than one isolated mode. The strongest constructed distribution has high alignment with the frontier-relevant modes, while the weakest has low contribution strength despite favorable outcome alignment.

The Appendix reports tests that vary the frontier construction inputs, together with

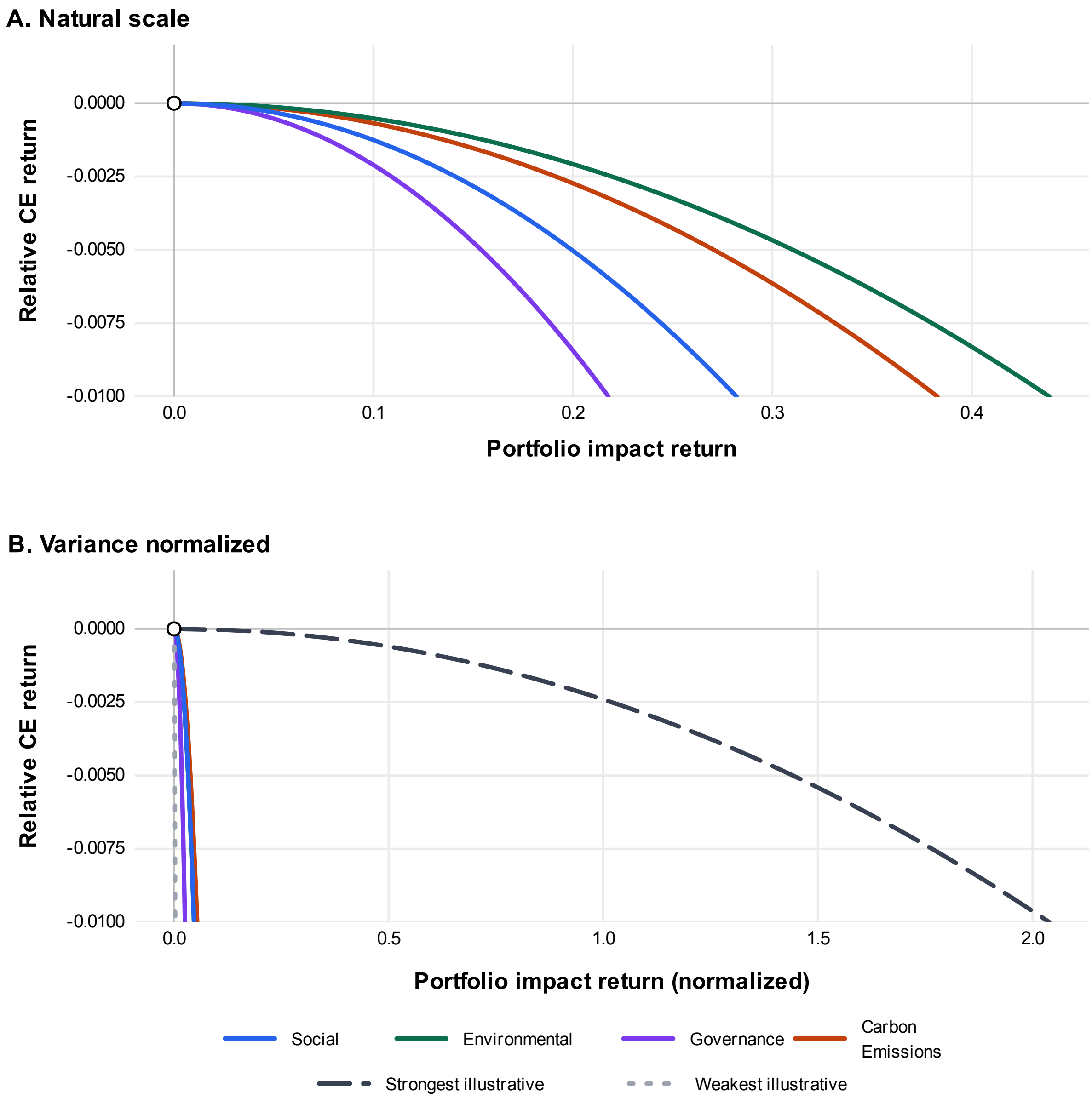

Figure 4: **Frontier scale across outcomes.** This figure reports Impact Frontiers for Aggregate-Supply CE and the full cross-section. Panel A uses Social, Environmental, Governance, and Carbon Emissions net outcome intensities on their natural scale; the x-axis reports portfolio impact return. Panel B first rescales each net outcome intensity direction to unit total-assets-weighted standard deviation so the outcome categories can be compared with the strongest and weakest illustrative distributions.

Outcome	$\log \eta^2$	$2S_g$	$2G_{\text{dist}}$	$2C$	B	$-R$	ϵ
Social	1.4	3.6	-1.0	-1.9	4.4	-3.5	-0.13
Environmental	2.3	4.5	-1.6	-1.7	4.3	-3.2	-0.04
Governance	0.9	4.3	-2.3	-2.0	4.7	-3.8	-0.09
Carbon Emissions	2.0	3.9	-1.0	-2.1	4.8	-3.6	-0.07
Strongest illustrative	5.3	0.0	13.5	-1.5	0.2	-6.9	0.00
Weakest illustrative	-7.9	0.0	8.3	-10.5	0.8	-6.5	-0.01

Table 3: **Frontier-strength decomposition for Figure 4.** The decomposition is $\log \eta^2 = 2S_g + 2G_{\text{dist}} + 2C + B - R + \epsilon$. S_g is the log standard deviation of the implemented net outcome intensity, G_{dist} is outcome-distribution alignment with frontier-relevant modes, C is contribution strength, B is modal breadth, R is CE-return cost, and ϵ is the modal-factorization residual. The two illustrative outcome distributions are normalized to unit standard deviation, so $S_g = 0$ for those rows. Their $\log \eta^2$ values are comparable with the observed rows on the normalized scale after subtracting $2S_g$ from the observed rows. Entries are rounded.

long-only, fixed-investment cases. The input tests move frontier scale in both directions: score transformations and score noise change the outcome scale, Market-Risk CE lowers tilt costs, and covariance regularization changes little. The long-only cases remain close to the unrestricted frontier. With the frontier construction fixed, the next subsection reports how much of each frontier the simple tilt rules recover.

6.2 Direct and net-outcome tilt recovery

Table 4 and Figure 5 compare impact-inefficient tilts to the frontier. The gap between the net-outcome and direct tilts isolates the recovery effect of direct contribution multipliers. The remaining gap between the direct tilt and the Impact Frontier isolates the effect of the full contribution multiplier matrix beyond those direct multipliers. In the full cross-section, direct tilts are close to the frontier for the observed outcome categories, so direct contribution multipliers explain most of the gain over the net-outcome tilt. Restricted active sets and constructed worst cases show that this need not hold generally.

Restricted active-set calculations use Market-Risk CE because they specialize to an investor whose active opportunity set is the restricted set: within that set, the investor bears the covariance risk of the available assets, but the calculation does not add the aggregate non-market-risk term $\mathbf{\Phi}_S^N$. In the reported SIC2 active-set cases, recovery remains positive in every observed outcome-by-SIC2 case. The wrong-direction results arise only in constructed worst cases, while observed net-outcome recovery can still fall to 0.17 in narrow active sets.

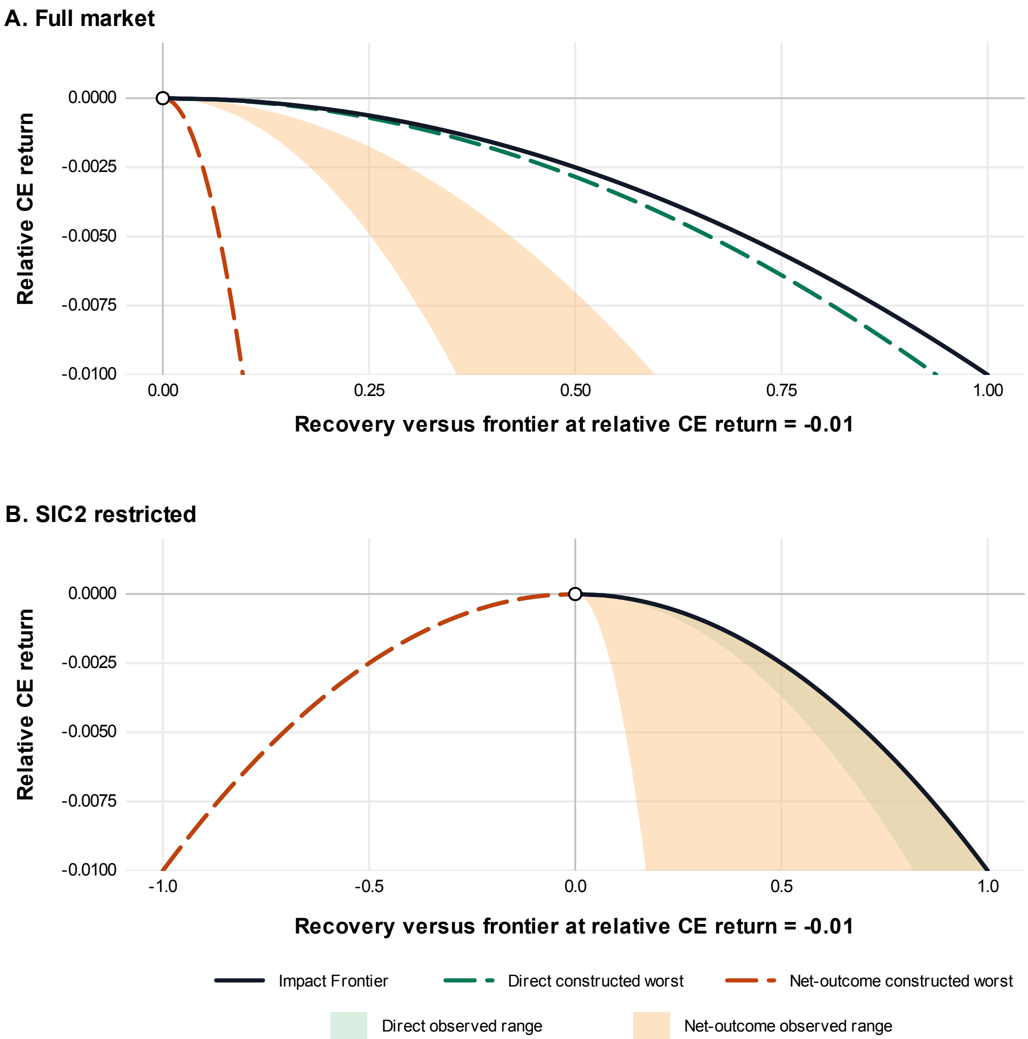

Figure 5: **Recovery relative to the Impact Frontier.** The figure reports the signed equal-CE recovery ratio $\tilde{\rho}$, normalized so that the Impact Frontier equals one at a CE-return cost of 100 basis points. Negative values indicate tilts that move portfolio impact return in the wrong direction. Panel A uses the full cross-section under Aggregate-Supply CE; shaded regions span the four outcome categories. Panel B uses SIC2-restricted active sets under Market-Risk CE; shaded regions span the outcome-category and SIC2 cases with at least five issuers. Dashed lines report constructed worst cases. Additional details are reported in Table 4.

In the baseline investor-supply specification for the full cross-section, observed direct-tilt recovery is 0.996–0.999, while net-outcome recovery is 0.36–0.60 (Table 4). The restricted market-risk-only specification produces the same ordering: direct-tilt recovery remains 0.975–0.991, while net-outcome recovery is 0.18–0.25. For constructed outcome-intensity distribu-

Setting	Tilt	Observed range	Lower endpoint	Upper endpoint	Worst-case recovery
Market	Direct	0.996 to 0.999	Carbon Emissions	Governance	0.938
Market	Net outcome	0.357 to 0.597	Environmental	Carbon Emissions	0.097
SIC2	Direct	0.821 to 1.000	Social, SIC2 58 (n=16)	Governance, SIC2 54 (n=5)	-1.000
SIC2	Net outcome	0.172 to 0.996	Environmental, SIC2 37 (n=41)	Governance, SIC2 17 (n=6)	-1.000

Table 4: **Recovery ranges in Figure 5.** The observed range column reports the lower and upper recovery range across the observed cases in each setting. The worst-case recovery column reports the constructed worst case plotted in Figure 5. Recoveries are signed equal-CE recovery ratios at a CE-return cost of 100 basis points. Negative values indicate tilts that move portfolio impact return in the wrong direction. Market rows use the full cross-section under Aggregate-Supply CE. SIC2 rows use the 2019 SIC2 active-set universe with at least five issuers, normalized implemented net outcome score directions, and Market-Risk CE, and report the outcome or SIC2 case defining each edge.

tions, restricted active sets can have negative recovery from both net-outcome and direct tilts.

While the frontier construction tests in the Appendix change frontier scale, the gap between the direct and net-outcome tilts remains economically large. The Appendix also provides analytic approximations to the constructed worst cases in the paper's outcome-intensity space. The dispersion in the direct contribution multipliers explains the 0.097 net-outcome minimum for the full cross-section. For the direct tilt, an approximation using the mean off-diagonal implies recovery of 0.92, close to the 0.94 constructed worst case.

7 Discussion

7.1 Interpretation and limitations

The baseline calibration keeps $\mathbf{\Phi}_D$ diagonal. This is a parsimonious restriction, not a claim that issuer capital demand is economically independent across firms. Input-output and production-network models provide natural motivations for off-diagonal issuer-demand terms because changes in one firm can propagate through customers, suppliers, and production linkages (Acemoglu et al. 2012; Herskovic 2018). Empirical evidence on customer-supplier and technological links suggests that such cross-firm connections are economically meaningful in asset markets as well (Cohen and Frazzini 2008; Lee et al. 2019).

The baseline calibration also keeps investor-specific trading constraints deliberately simple. Limits on position size, leverage, borrowing, or short sales can be imposed, but the specifics vary by investor and are not central to the mechanism.

The calibration does not separately identify primary issuance, secondary-market trading, repurchases, or temporary trading paths, and it abstracts from issuer capital structure. The Sharpe-ratio representation follows Betermier et al. (2026) and reduces leverage sensitivity when corporate debt is close to risk-free. It does not, however, separate debt and equity financing channels. Extending the model to multiple claims per issuer, with different payoff rights, pledgeability, tax shields, and financing uses, is an important direction for future research.

The model is not specific to public equity. It can be applied to private markets, debt markets, or other financing channels when market-specific measures of capital supply, issuer capital, prices, and outcome intensities are available. This matters because impact-investing discussions often treat private markets as the natural setting for positive capital-allocation impact (Brest et al. 2018). In the model, that requires low investor-side supply elasticity, low issuer-demand resistance, or both. Private markets may indeed satisfy those conditions when investor participation is thinner or financing constraints bind. Yet this paper shows that public equities need not fail them.

Estimating marginal outcome intensities is its own scientific, economic, and normative problem. A practical challenge is that standard accounting metrics and other familiar firm-level information are not marginal values. This matters because impact on the margin can differ sharply from the average effects attributed to firms (Hartzmark and Shue 2022). The same marginal-accounting issue applies to the risk-free asset: $\boldsymbol{g}_{\text{net}}$ depends on $\bar{g}_B$, the outcome intensity assigned to the risk-free-asset holdings reduced when issuer capital rises. Allcott et al. (2026) provide a leading example of marginal impact modeling, constructing corporate social-impact estimates across consumer welfare, worker surplus, and environmental externalities. In other applications, more abstract outcome-intensity proxies could be used, such as estimates of how marginal capital contributes to sustainability-transition pathways (Penna et al. 2023). Such applications may also require richer supply-side specifications to capture complex investor networks (Kartun-Giles and Ameli 2023).

Contribution multipliers provide a marginal version of additionality, an important concept in development finance (Carter et al. 2021). Additionality is often framed as a binary question about whether an outcome would have occurred without the investment. That binary framing is too coarse for marginal investors, and investor impact is continuous in this model. A capital-supply shift can be absorbed by other investors, become issuer capital, and be offset or amplified through the system response. The question is not whether an investment is additional, but how much portfolio impact return it produces.

The estimates here are marginal capital-allocation statistics for public equities, not a dynamic welfare analysis. The calibration does not endogenize investor consumption, saving, labor income, wealth shares, return volatilities and correlations, or capital adjustment costs. These features are central in dynamic production models such as Favilukis et al. (2025) and are natural directions for future work.

7.2 Impact returns and cost-effectiveness

The frontier is a marginal cost-effectiveness curve for the capital-allocation channel. The relevant comparison is the investor's comparative advantage in producing marginal outcome change through capital allocation relative to engagement, stewardship, direct investment, philanthropy, policy advocacy, and other channels. This comparison parallels marginal project appraisal in imperfect economies and philanthropic cause prioritization (Drèze and Stern 1987; Oehlsen 2024). Those traditions often combine an empirical outcome estimate with a shadow price or welfare weight, which this paper leaves to the investor or analyst. The scale of outcomes, tractability of solutions, neglectedness by other funders, and room for more funding all have analogues in the model, including in the contribution multipliers, and all such analyses involve substantial uncertainty. The multi-asset setting also adds financial risk and the system response, which are not typically part of project-level grant analysis.

High contribution multipliers do not by themselves imply low marginal cost per unit of outcome change. A portfolio tilt can use only the cross-sectional variation available among investable firms, and that variation may be weakly aligned with the target outcome. As an illustrative example for Carbon Emissions, Table 5 converts the variance-normalized Carbon Emissions Impact Frontier from Figure 4, Panel B, into dollars per ton. The calculation reports illustrative conversions in which one Carbon Emissions score standard deviation corresponds to 100 or 1000 tons of CO_2e per million dollars of issuer capital. These round scales are motivated by the order-of-magnitude range of the emissions-intensity standard deviations per million dollars of revenue reported by Bolton and Kacperczyk (2021); neither conversion is an estimate of emissions per dollar of issuer capital or realized emissions abatement. The frontier gets steeper as portfolio impact return and CE-return cost increase, increasing marginal costs per ton, even though the contribution multipliers remain the same.

		Marginal cost	
CE-return cost	Portfolio impact return (score s.d.)	100 tons per \$1 million	1,000 tons per \$1 million
1 bp	0.006	\$361 per ton	\$36.1 per ton
10 bps	0.017	\$1143 per ton	\$114 per ton
100 bps	0.055	\$3615 per ton	\$361 per ton

Table 5: **Illustrative cost-effectiveness on the Impact Frontier.** The table converts the baseline Carbon Emissions Impact Frontier into implied dollars per ton of marginal investor CO_2e impact under illustrative conversion rates. The conversions assume that one Carbon Emissions score standard deviation corresponds to 100 or 1000 tons of CO_2e per million dollars of issuer capital. The Carbon Emissions score is oriented so that positive outcome change corresponds to lower emissions. Portfolio impact return reports the expected score-standard-deviation issuer-capital change per unit investor wealth on the Impact Frontier at the stated CE-return cost. Marginal cost reports the carbon price that would equate the marginal CE-return cost on the frontier with one additional ton under each illustrative conversion. The dollar-per-ton values depend on the illustrative score-to-tons conversion and the frontier calibration; they are not estimates of a particular investor's CE cost or realized emissions abatement.

7.3 Valuing impact

Figure 6 shows the Impact Frontier for an equal-weight blended outcome score across Social, Environmental, Governance, and Carbon Emissions. The outcome mix shows how equal blended weights lead to different movements in the underlying outcome categories. Equal weights do not imply equal movements in each category because the capital-allocation system changes the cost of moving each outcome. The mix is governed by the outcome-effectiveness matrix defined in the Appendix. For a specified values vector over outcome categories, the matrix determines which outcome directions the capital-allocation channel moves most effectively.

The values vector $\boldsymbol{\nu}$ is an investor input rather than an econometric output. Its entries can represent willingness to pay for each outcome, marginal costs in other channels, or institutional hurdle values. These possibilities are related to the value-versus-values framing in Starks (2023), which emphasizes that sustainable-finance choices can reflect both financial value and broader investor values. The Impact Frontier then translates that input into the outcome mix produced by the calibrated capital-allocation channel; a balanced values vector need not produce balanced outcome changes.

Two practical frictions sit outside the frontier calculation. First, the outcome-intensity vector is measured with imperfect sustainability information because reporting varies in scope, materiality, implementation, and enforcement (Christensen et al. 2021). The frontier construction can accommodate different values vectors and outcome-intensity inputs as information and preferences change. Second, delegated investors must choose whose preferences determine $\boldsymbol{\nu}$ and which procedure aggregates them. Private investors can choose this vector directly. Delegated institutions face a governance problem: which beneficiaries' preferences control $\boldsymbol{\nu}$, and by which procedure? Recent work on shareholder and investor democracy studies mechanisms for eliciting and aggregating investor or beneficiary preferences (Hart et al. 2025; Bauer et al. 2026).

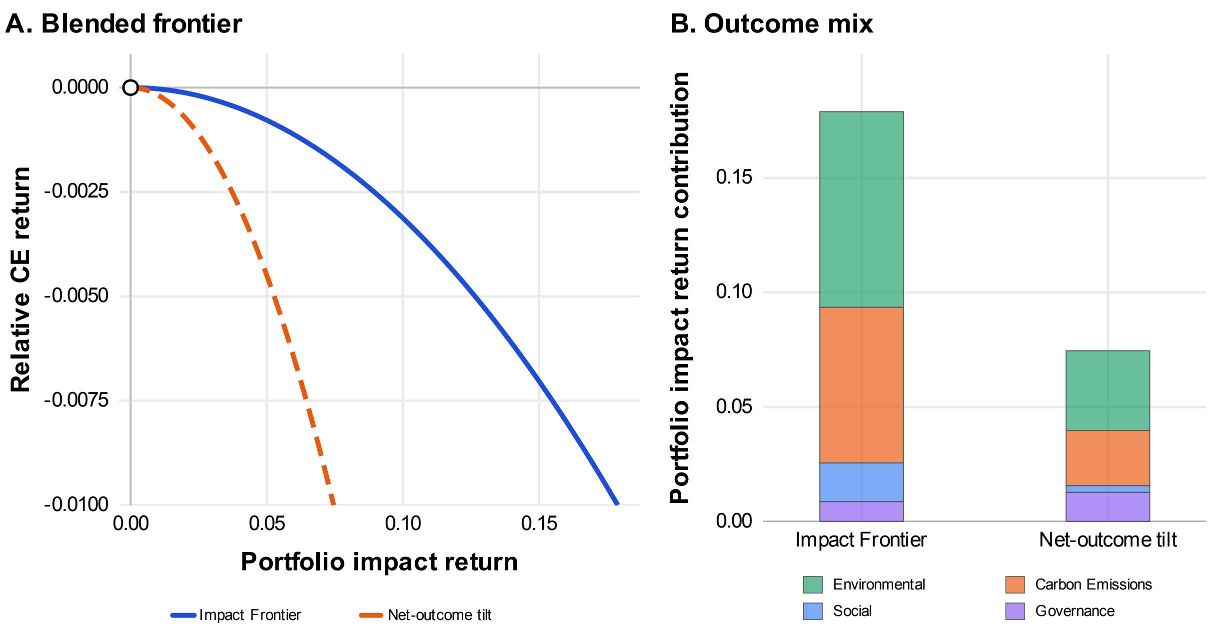

Figure 6: **Blended frontier and outcome mix.** Panel A compares the Impact Frontier with the net-outcome tilt for the equal-weight blended outcome score using the baseline calibration under Aggregate-Supply CE. Panel B decomposes the outcome mix at a relative CE return of −100 basis points. All four outcome categories are oriented so positive change is favorable.

8 Conclusion

How much can a sustained, marginal portfolio tilt by an investor change firms' capital and the external outcomes those firms produce? This paper's answer is the Impact Frontier, the Pareto frontier between certainty-equivalent returns and portfolio impact returns.

The theory identifies the supply-and-demand system, outcome accounting, and the effective covariance matrix as the empirical inputs that determine frontier strength and portfolio-tilt performance. Contribution multipliers connect a supply-schedule shift to equilibrium issuer capital; outcome accounting converts those capital changes into impact returns. Raw outcome intensities attribute outcomes to firms, whereas impact returns represent expected investor contribution in equilibrium. In the U.S. public-equity calibration, tilts that account for contribution-multiplier heterogeneity produce impact returns close to the frontier, while tilts based only on the outcomes attributed to firms can fall far below. Direct contribution multipliers explain most of the full-cross-section gain over net-outcome tilts; restricted active sets and constructed worst cases show where the full contribution multiplier matrix becomes more important.

The frontier provides a marginal cost curve for comparing capital allocation with other channels of investor impact. Its scale and shape depend on the empirical inputs. The broader point is general: investor impact is an equilibrium quantity to be estimated, not assumed away or asserted by intention.

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Online Appendix for
The Impact Frontier

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This appendix provides derivations for the theoretical results in the paper, details the data construction and empirical methodology, and reports additional results and discussion.

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OA.1 Local equilibrium and the risk-free asset

This section derives the local equilibrium response behind the contribution multiplier matrix and the net outcome intensity. The derivation is first order around the benchmark market-clearing point. The marginal investor is small in the sense that the surrounding schedules and slope matrices are held fixed, not in the sense that the derivative is zero.

The paper uses the baseline risk-free-asset accounting $\mathrm{d}B = -\mathbf{1}'\mathrm{d}\boldsymbol{K}$ and holds the risk-free rate fixed. This appendix allows a general cash requirement $\boldsymbol{\Pi}'\mathrm{d}\boldsymbol{K}$, with $\boldsymbol{\Pi} = \mathbf{1}$ in the baseline, and records both polar cases for risk-free-asset supply: the highly elastic case that holds the risk-free rate fixed and yields the baseline contribution multiplier matrix, and the inelastic case in which the risk-free rate adjusts.

OA.1.A Local system with a risk-free asset

Let $\boldsymbol{P}$ denote risky-asset valuation variables, let R_f denote the gross risk-free rate, and let $\boldsymbol{K}_D(\boldsymbol{P}, R_f)$ and $\boldsymbol{K}_S(\boldsymbol{P}, R_f)$ denote issuer demand and aggregate investor supply for issuer capital. Let B denote aggregate risk-free-asset holdings by the investor sector. Let $\boldsymbol{\Pi} \in \mathbb{R}^{N_A}$ denote the cash absorbed per unit increase in issuer capital, so

$$\mathrm{d}B = -\boldsymbol{\Pi}'\mathrm{d}\boldsymbol{K}_D. \tag{OA.1}$$

Evaluate derivatives at the benchmark equilibrium and define

$$\boldsymbol{\Xi}_D \equiv \left.\frac{\partial \boldsymbol{K}_D}{\partial \boldsymbol{P}}\right|_*, \qquad \boldsymbol{\Xi}_S \equiv -\left.\frac{\partial \boldsymbol{K}_S}{\partial \boldsymbol{P}}\right|_*, \qquad \boldsymbol{\Xi}_T \equiv \boldsymbol{\Xi}_D + \boldsymbol{\Xi}_S, \tag{OA.2}$$

$$\boldsymbol{a}_D \equiv \left.\frac{\partial \boldsymbol{K}_D}{\partial R_f}\right|_*, \qquad \boldsymbol{a}_S \equiv \left.\frac{\partial \boldsymbol{K}_S}{\partial R_f}\right|_*, \qquad \boldsymbol{a}_T \equiv \boldsymbol{a}_D - \boldsymbol{a}_S. \tag{OA.3}$$

When $\boldsymbol{\Xi}_D$ and $\boldsymbol{\Xi}_S$ are nonsingular, define

$$\boldsymbol{\Phi}_D \equiv \boldsymbol{\Xi}_D^{-1}, \qquad \boldsymbol{\Phi}_S \equiv \boldsymbol{\Xi}_S^{-1}, \qquad \boldsymbol{\mathcal{C}} \equiv \boldsymbol{\Xi}_D\boldsymbol{\Xi}_T^{-1} = (\boldsymbol{\Phi}_D + \boldsymbol{\Phi}_S)^{-1}\boldsymbol{\Phi}_S. \tag{OA.4}$$

A marginal supply shift by investor i enters aggregate investor supply as $\boldsymbol{I}_i'\delta\boldsymbol{k}_i$. Risky-market clearing implies

$$\boldsymbol{\Xi}_T \mathrm{d}\boldsymbol{P} + \boldsymbol{a}_T \mathrm{d}R_f = \boldsymbol{I}_i' \delta \boldsymbol{k}_i. \tag{OA.5}$$

Issuer capital changes by

$$\mathrm{d}\boldsymbol{K}_D = \boldsymbol{\mathcal{C}} \boldsymbol{I}_i' \delta \boldsymbol{k}_i + \boldsymbol{\ell}_f \mathrm{d}R_f, \tag{OA.6}$$

where $\boldsymbol{\ell}_f = \boldsymbol{a}_D - \boldsymbol{\mathcal{C}} \boldsymbol{a}_T$.

OA.1.B Risk-free-asset supply and net outcome intensity

If risk-free-asset supply has local slope κ_f, then $\mathrm{d}B = \kappa_f \mathrm{d}R_f$. Combining this with (OA.1) implies

$$\kappa_f \mathrm{d}R_f + \boldsymbol{\Pi}' \mathrm{d}\boldsymbol{K}_D = 0. \tag{OA.7}$$

Combining (OA.6) and (OA.7), define

$$S_f \equiv \kappa_f + \boldsymbol{\Pi}' \boldsymbol{\ell}_f. \tag{OA.8}$$

If $S_f \neq 0$, then

$$\mathrm{d}R_f = -S_f^{-1} \boldsymbol{\Pi}' \boldsymbol{\mathcal{C}} \boldsymbol{I}_i' \delta \boldsymbol{k}_i, \tag{OA.9}$$

$$\mathrm{d}\boldsymbol{K}_D = \left(\boldsymbol{\mathcal{C}} - \boldsymbol{\ell}_f S_f^{-1} \boldsymbol{\Pi}' \boldsymbol{\mathcal{C}} \right) \boldsymbol{I}_i' \delta \boldsymbol{k}_i. \tag{OA.10}$$

If risk-free-asset supply is highly elastic, $\kappa_f \to \infty$, then $\mathrm{d}R_f \to 0$ and

$$\mathrm{d}\boldsymbol{K}_D = \boldsymbol{\mathcal{C}} \boldsymbol{I}_i' \delta \boldsymbol{k}_i. \tag{OA.11}$$

If the risk-free asset has outcome intensity $\bar{g}_B$, the outcome change is

$$\Delta X_i = (\boldsymbol{g} - \bar{g}_B \boldsymbol{\Pi})' \boldsymbol{\mathcal{C}} \boldsymbol{I}_i' \delta \boldsymbol{k}_i. \tag{OA.12}$$

If risk-free-asset supply is locally inelastic, $\kappa_f = 0$, then $\mathbf{\Pi}'\mathrm{d}\boldsymbol{K}_D = 0$ and

$$\mathrm{d}\boldsymbol{K}_D = \left(\boldsymbol{I} - \boldsymbol{\ell}_f(\mathbf{\Pi}'\boldsymbol{\ell}_f)^{-1}\mathbf{\Pi}'\right)\boldsymbol{\mathcal{C}}\boldsymbol{I}_i'\delta\boldsymbol{k}_i, \tag{OA.13}$$

provided $\mathbf{\Pi}'\boldsymbol{\ell}_f \neq 0$. Defining

$$\bar{g}_B \equiv \frac{\boldsymbol{g}'\boldsymbol{\ell}_f}{\mathbf{\Pi}'\boldsymbol{\ell}_f} \tag{OA.14}$$

produces

$$\boldsymbol{g}'\mathrm{d}\boldsymbol{K}_D = (\boldsymbol{g} - \bar{g}_B\mathbf{\Pi})'\boldsymbol{\mathcal{C}}\boldsymbol{I}_i'\delta\boldsymbol{k}_i. \tag{OA.15}$$

Thus both polar cases use the same downstream impact-return expression after replacing gross outcome intensities by $\boldsymbol{g}_{\text{net}} = \boldsymbol{g} - \bar{g}_B\mathbf{\Pi}$. The empirical baseline does not estimate $\boldsymbol{\ell}_f$. It uses the baseline accounting $\mathbf{\Pi} = \mathbf{1}$ and sets $\bar{g}_B$ equal to the total-assets-weighted mean gross outcome intensity.

OA.2 Aggregate investor supply slope matrix

Write investor i's marginally active holdings as $\boldsymbol{k}_i \in \mathbb{R}^{|\mathcal{N}_i|}$. The matrix $\boldsymbol{I}_i$ selects investor i's investable assets from the full asset vector. Investors pinned at lower or upper position limits affect aggregate levels and market clearing, but not the local aggregate supply slope. Their fixed holdings are absorbed into the equilibrium level and do not enter the slope derivation.

A local affine investor supply schedule is written as $\boldsymbol{I}_i\boldsymbol{P} = \boldsymbol{\phi}_i - \mathbf{\Phi}_{ii}\boldsymbol{k}_i$. This implies $\boldsymbol{k}_i = \mathbf{\Phi}_{ii}^{-1}\Big(\boldsymbol{\phi}_i - \boldsymbol{I}_i\boldsymbol{P}\Big)$. Unless stated otherwise, all slope parameters in this section are evaluated at the same equilibrium point used for the local affine approximation. The vector $\boldsymbol{\phi}_i$ is the local residual term that makes the affine approximation pass through the schedule at $(\boldsymbol{P}^*, \boldsymbol{K}^*)$.

Aggregate capital supplied by the marginally active investor base is

$$\boldsymbol{K} = \sum_i \boldsymbol{I}_i'\boldsymbol{k}_i = \sum_i \boldsymbol{I}_i'\mathbf{\Phi}_{ii}^{-1}\Big(\boldsymbol{\phi}_i - \boldsymbol{I}_i\boldsymbol{P}\Big).$$

Assuming the aggregate precision matrix is nonsingular, rearranging yields the aggregate reduced-form supply schedule

$$\boldsymbol{P} = \boldsymbol{\phi}_S - \boldsymbol{\Phi}_S \boldsymbol{K} \tag{OA.16}$$

with

$$\boldsymbol{\Phi}_S \equiv \left(\sum_i \boldsymbol{I}_i' \boldsymbol{\Phi}_{ii}^{-1} \boldsymbol{I}_i \right)^{-1}, \qquad \boldsymbol{\phi}_S \equiv \boldsymbol{\Phi}_S \left(\sum_i \boldsymbol{I}_i' \boldsymbol{\Phi}_{ii}^{-1} \boldsymbol{\phi}_i \right). \tag{OA.17}$$

A useful special case is if $I_i = I$ for all active investors and $\boldsymbol{\Phi}_{ii} = s_i \boldsymbol{\Phi}_0$ with $s_i > 0$, then

$$\boldsymbol{\Phi}_S = \left(\sum_i s_i^{-1} \right)^{-1} \boldsymbol{\Phi}_0.$$

Thus proportional local supply slopes aggregate to the same slope matrix up to scale. This case motivates using the Aggregate-Supply CE specification to study full cross-section tilts without imposing arbitrary active-set constraints.

OA.2.A One-factor approximation for $\boldsymbol{\Phi}_S$

Suppose the horizon payoff covariance has the one-factor form

$$\boldsymbol{\Sigma}_{\text{val}} = \boldsymbol{b}\boldsymbol{b}' + \operatorname{diag}(\boldsymbol{\sigma}_\epsilon^2) = \operatorname{diag}(\boldsymbol{\sigma}_{\text{val}})(\boldsymbol{\rho}\boldsymbol{\rho}' + \operatorname{diag}(\mathbf{1} - \boldsymbol{\rho}^2)) \operatorname{diag}(\boldsymbol{\sigma}_{\text{val}}), \tag{OA.18}$$

where $\boldsymbol{b}$ is exposure to the common payoff factor, $\sigma_{\epsilon,n}^2$ is idiosyncratic payoff variance, $\boldsymbol{\sigma}_{\text{val}}$ is overall asset volatility, and $\boldsymbol{\rho}$ is the correlation with the common factor, which has unit variance. Assume that investor i's local slope matrix is $\boldsymbol{\Phi}_{ii} = \gamma_i \boldsymbol{I}_i \boldsymbol{\Sigma}_{\text{val}} \boldsymbol{I}_i'$, where γ_i is marginal risk aversion in dollar-holding units. In the standard mean-variance case, $\gamma_i = \gamma_i^{\text{rr}} / W_i$, where γ_i^{rr} is relative risk aversion and W_i is investor wealth. Let

$$\boldsymbol{D}_\epsilon \equiv \operatorname{diag}(\boldsymbol{\sigma}_\epsilon^2), \qquad \bar{\gamma} \equiv \left(\sum_i \gamma_i^{-1} \right)^{-1}, \qquad q_n \equiv \bar{\gamma} \sum_i \gamma_i^{-1} \mathbf{1}_{\{n \in \mathcal{N}_i\}}.$$

Thus $\bar{\gamma}$ is aggregate marginal risk aversion in dollar-holding units, and q_n is the active-investor share for asset n, weighted by γ_i^{-1}. Applying the aggregate supply slope formula in (OA.17) and replacing the investor-specific active-set matrices in the common-factor correction by their weighted-average active-set term yields

$$\Phi_S \approx \bar{\gamma}\left(\boldsymbol{b}\boldsymbol{b}' + \operatorname{diag}(\boldsymbol{q})^{-1}\boldsymbol{D}_\epsilon\right) \tag{OA.19}$$

$$= \bar{\gamma}\boldsymbol{\Sigma}_{\text{val}} + \bar{\gamma}\operatorname{diag}\left(\boldsymbol{q}^{-1} - \mathbf{1}\right)\operatorname{diag}\left(1 - \boldsymbol{\rho}^2\right)\operatorname{diag}(\boldsymbol{\sigma}^2_{\text{val}}). \tag{OA.20}$$

Merton (1987) assumes all investors have access to a forward contract on the common factor; that asset removes the active-set-specific common-factor correction and makes the corresponding expression exact. Without that asset, (OA.20) is an approximation: it is exact under common active sets and accurate when active-set dispersion is small relative to the common-factor loading. The useful point of the one-factor approximation is the decomposition of aggregate supply. The common factor generates off-diagonal structure because market payoff risk links assets in the aggregate supply schedule. The active-set term adds an asset-specific non-market margin: when the effective active investor base q_n is smaller, residual risk is costlier for the marginal investor base to absorb. The market-risk-only frontier sensitivities in Appendix Table OA.11 use the full regularized correlation matrix, while this rank-one analysis isolates the same economic mechanism.

OA.2.B Scaled return representations

The empirical system is estimated with Sharpe ratios on the left-hand side: the investor-side equation uses the market-risk and non-market-risk terms, while the issuer-side equation uses log total-assets growth. This section reports the scaled representations needed to express market portfolios, frontier portfolios, and comparison tilts on a common scale. The three representations are μ-V, λ-Q, and market-relative λ-ρ; these are changes in units around the observed market point, not changes in economics.

The scaling chain is: $\boldsymbol{V} \approx \operatorname{diag}(\boldsymbol{P}^*)\boldsymbol{K}, \boldsymbol{Q} \approx \operatorname{diag}(\boldsymbol{\sigma}^*_R)\boldsymbol{V}, \boldsymbol{\rho}^{\text{int}}_M \approx (Q^*_M)^{-1}\boldsymbol{Q}$. Combining this with the affine aggregate supply schedule (OA.16) produces the local excess-return representation. Write $\boldsymbol{\phi}^\mu_S$ for the inherited excess-return intercept, including subtraction of the risk-free rate, and write $\mathbf{\Phi}_S^V \equiv \operatorname{diag}(\boldsymbol{P}^*)^{-1}\mathbf{\Phi}_S \operatorname{diag}(\boldsymbol{P}^*)^{-1}$. Then $\boldsymbol{\mu} \equiv \mathbb{E}[\boldsymbol{r}] - r_f\mathbf{1} = \boldsymbol{\phi}_S^\mu + \mathbf{\Phi}_S^V\boldsymbol{V}$. Volatility scaling implies $\boldsymbol{\lambda} = \boldsymbol{\phi}_S^\lambda + \mathbf{\Phi}_S^Q\boldsymbol{Q}$, where $\boldsymbol{\phi}_S^\lambda = \operatorname{diag}(\boldsymbol{\sigma}_R^*)^{-1}\boldsymbol{\phi}_S^\mu$ and $\mathbf{\Phi}_S^Q = \operatorname{diag}(\boldsymbol{\sigma}_R^*)^{-1}\mathbf{\Phi}_S^V \operatorname{diag}(\boldsymbol{\sigma}_R^*)^{-1}$. Holding market risk capital fixed at Q_M^* yields the market-relative representation $\boldsymbol{\lambda} = \boldsymbol{\phi}_S^\lambda + \mathbf{\Phi}_S^\rho\boldsymbol{\rho}_M^{\text{int}}$, with $\mathbf{\Phi}_S^\rho = \mathbf{\Phi}_S^Q Q_M^*$. The issuer side admits the same twin representation with the sign reversal appropriate to a demand schedule.

OA.2.C Regularized covariance and aggregate supply matrices

The frontier calculations use a regularized covariance matrix and the corresponding aggregate supply matrix.

Let $\hat{\mathbf{\Sigma}}_t^F$ be the annualized forward-window covariance matrix and let $\hat{\boldsymbol{w}}_{M,t}$ be market weights. The financial layer separates the forward-window covariance matrix into a market-factor part and a residual covariance matrix, then shrinks the residual covariance matrix:

$$\hat{\mathbf{\Sigma}}_t^* = \hat{\boldsymbol{\beta}}_{M,t}^F\hat{\boldsymbol{\beta}}_{M,t}^{F\prime}\hat{\sigma}_{M,t}^{2,F} + \omega_t\frac{\operatorname{tr}(\hat{\boldsymbol{V}}_{e,t}^{data})}{N_t}\boldsymbol{I} + (1-\omega_t)\hat{\boldsymbol{V}}_{e,t}^{data}, \qquad \omega_t = \frac{N_t}{N_t + T_t},$$

where $\hat{\sigma}_{M,t}^{2,F} = \hat{\boldsymbol{w}}_{M,t}'\hat{\mathbf{\Sigma}}_t^F\hat{\boldsymbol{w}}_{M,t}$, $\hat{\boldsymbol{\beta}}_{M,t}^F = \hat{\mathbf{\Sigma}}_t^F\hat{\boldsymbol{w}}_{M,t}/\hat{\sigma}_{M,t}^{2,F}$, and $\hat{\boldsymbol{V}}_{e,t}^{data} = \hat{\mathbf{\Sigma}}_t^F - \hat{\boldsymbol{\beta}}_{M,t}^F\hat{\boldsymbol{\beta}}_{M,t}^{F\prime}\hat{\sigma}_{M,t}^{2,F}$. Let $\hat{\boldsymbol{D}}_{\sigma,t} = \operatorname{diag}(\sqrt{\operatorname{diag}(\hat{\mathbf{\Sigma}}_t^*)})$ and $\hat{\boldsymbol{R}}_t^* = \hat{\boldsymbol{D}}_{\sigma,t}^{-1}\hat{\mathbf{\Sigma}}_t^*\hat{\boldsymbol{D}}_{\sigma,t}^{-1}$. For each empirical calibration scenario s, the λ-ρ^{int} aggregate supply slope matrix is

$$\hat{\mathbf{\Phi}}_{S,s}^\rho = \hat{\mathbf{\Phi}}_{S,s}^{M,\rho} + \hat{\mathbf{\Phi}}_{S,s}^{N,\rho}, \qquad \hat{\mathbf{\Phi}}_{S,s}^{M,\rho} = \hat{\Delta}_{M,s}\hat{\boldsymbol{R}}_s^*, \qquad \hat{\mathbf{\Phi}}_{S,s}^{N,\rho} = \operatorname{diag}(\hat{\boldsymbol{d}}_{S,s}^N).$$

The diagonal entries are

$$\hat{d}_{S,n,s}^N = \hat{\Delta}_{N,s}(1-\hat{p}_{n,s}^{TA})(1-\hat{\rho}_{M,n,s}^2).$$

The product of $\hat{d}_{S,n,s}^N$ and $\hat{\rho}_{M,n,s}^{\text{int}}$ is the fitted non-market-risk term in the Sharpe-ratio equation. The corresponding matrices in Q and market-value units are

$$\hat{\mathbf{\Phi}}_{S,s}^Q = \hat{\mathbf{\Phi}}_{S,s}^\rho/\hat{Q}_{M,s}, \qquad \hat{\mathbf{\Phi}}_{S,s}^V = \hat{\boldsymbol{D}}_{\sigma,s}\hat{\mathbf{\Phi}}_{S,s}^Q\hat{\boldsymbol{D}}_{\sigma,s}.$$

Here s indexes the fitted specification and panel-year calibration.

OA.2.D Sharpe-ratio representation of aggregate supply

The local aggregate investor-supply schedule may also be expressed with expected excess returns on the left. The intercept is inherited from the affine supply schedule after benchmark-price scaling and subtraction of the risk-free rate; it absorbs the benchmark return level and all required-return components not represented by the local slope terms. With $\boldsymbol{\mu} \equiv \mathbb{E}[\boldsymbol{r}] - r_f \mathbf{1}$,

$$\boldsymbol{\mu} = \boldsymbol{\Phi}_S^V \boldsymbol{V} + \boldsymbol{\phi}_S^\mu. \tag{OA.21}$$

The vector $\boldsymbol{\phi}_S^\mu$ is the supply intercept in expected-excess-return units. Its Sharpe-ratio counterpart is $\boldsymbol{\phi}_S^\lambda = \operatorname{diag}(\boldsymbol{\sigma}_R^*)^{-1} \boldsymbol{\phi}_S^\mu$.

Let $\boldsymbol{w}_M = \boldsymbol{V}/V_M$, let $\boldsymbol{\Sigma}_R$ be the risky-return covariance matrix, and define

$$\sigma_M^2 = \boldsymbol{w}_M' \boldsymbol{\Sigma}_R \boldsymbol{w}_M, \qquad \boldsymbol{\beta}_M = \frac{\boldsymbol{\Sigma}_R \boldsymbol{w}_M}{\sigma_M^2}. \tag{OA.22}$$

Add and subtract the CAPM market-risk term in the same μ-V units:

$$\boldsymbol{\mu} = \bar{\gamma} \boldsymbol{\Sigma}_R \boldsymbol{V} + \boldsymbol{\phi}_S^\mu + \boldsymbol{\delta}_{S,\Phi}^\mu, \qquad \text{with } \boldsymbol{\delta}_{S,\Phi}^\mu \equiv V_M \left(\boldsymbol{\Phi}_S^V - \bar{\gamma} \boldsymbol{\Sigma}_R \right) \boldsymbol{w}_M. \tag{OA.23}$$

Since $\boldsymbol{\Sigma}_R \boldsymbol{V} = V_M \sigma_M^2 \boldsymbol{\beta}_M$, the first term is the CAPM market-risk term. Converting to Sharpe-ratio units yields

$$\boldsymbol{\lambda} = \lambda_M \boldsymbol{\rho}_M + \boldsymbol{\phi}_S^\lambda + \boldsymbol{\delta}_{S,\Phi}^\lambda, \tag{OA.24}$$

$$\lambda_M \equiv \frac{\mathbb{E}[r_M] - r_f}{\sigma_M}, \qquad \boldsymbol{\delta}_{S,\Phi}^\lambda = \operatorname{diag}(\boldsymbol{\sigma}_R^*)^{-1} \boldsymbol{\delta}_{S,\Phi}^\mu. \tag{OA.25}$$

Inserting the one-factor aggregate supply-slope expression in (8) into (OA.25) yields

$$\boldsymbol{\delta}_{S,\Phi}^\lambda = \bar{\gamma} \operatorname{diag}\left(\boldsymbol{q}^{-1} - \mathbf{1}\right) \operatorname{diag}\left(1 - \boldsymbol{\rho}_M^2\right) \boldsymbol{Q} \tag{OA.26}$$

$$= \bar{\gamma} Q_M \operatorname{diag}\left(\boldsymbol{q}^{-1} - \mathbf{1}\right) \operatorname{diag}\left(1 - \boldsymbol{\rho}_M^2\right) \boldsymbol{\rho}_M^{\text{int}}. \tag{OA.27}$$

Taking the local differential of (OA.24), holding $\boldsymbol{\phi}_S^\lambda$ fixed, yields the vector form of (9). In the one-factor aggregate-supply approximation, the market-risk coefficient is $\bar{\gamma}Q_M$, and the non-market-risk component is (OA.27).

OA.3 Frontier strength and outcome scaling

This section collects the outcome-scaling and frontier-strength calculations. The first subsection describes the construction of net outcome intensities, and the second subsection decomposes frontier strength into outcome-intensity scale, outcome-distribution alignment, contribution strength, modal breadth, and CE-return cost.

OA.3.A Risk-free-asset outcome-intensity sensitivity

The baseline public-equity calculation sets $\mathbf{\Pi} = \mathbf{1}$ and assigns $\bar{g}_B$ to the total-assets-weighted mean gross outcome intensity. The net outcome intensity is

$$\boldsymbol{g}_{\text{net}}(\bar{g}_B) = \boldsymbol{g} - \bar{g}_B\mathbf{1}.$$

In the empirical frontier scripts, missing entries of a raw MSCI score are filled with that score's total-assets-weighted mean before this subtraction, so the corresponding net outcome intensity is zero.

Alternative values of $\bar{g}_B$ leave the estimated supply and demand slope matrices unchanged. Under the baseline $\mathbf{\Pi} = \mathbf{1}$,

$$\boldsymbol{h}(\bar{g}_B) = \boldsymbol{\mathcal{C}}'(\boldsymbol{g} - \bar{g}_B\mathbf{1}) = \boldsymbol{\mathcal{C}}'\boldsymbol{g} - \bar{g}_B\boldsymbol{\mathcal{C}}'\mathbf{1}.$$

Thus the sensitivity to $\bar{g}_B$ is one-dimensional once $\boldsymbol{\mathcal{C}}$ is fixed.

Table OA.1 reports Aggregate-Supply CE frontier strength and equal-CE tilt recovery for the Social outcome category across alternative values of $\bar{g}_B$. The direct tilt remains close to the frontier across the grid. The net-outcome tilt recovery changes in level but remains well below the direct tilt.

$\bar{g}_B$ setting	$\bar{g}_B$	Frontier strength	Direct recovery	Net-outcome recovery
Baseline total-assets mean	4.4	4.0	1.0	0.48
Minimum raw score	0.20	30	0.86	0.37
Raw-score 5th percentile	2.3	9.0	0.89	0.49
Raw-score 25th percentile	3.7	3.9	0.98	0.52
Median raw score	4.5	4.2	0.99	0.47
Raw-score 75th percentile	5.4	7.4	0.95	0.43
Raw-score 95th percentile	6.9	19	0.90	0.39
Maximum raw score	10	70	0.87	0.36

Table OA.1: **Risk-free-asset outcome-intensity sensitivity.** The table varies $\bar{g}_B$, recomputes $\boldsymbol{g}_{\text{net}}(\bar{g}_B)$ and $\boldsymbol{h}(\bar{g}_B) = \boldsymbol{\mathcal{C}}'\boldsymbol{g}_{\text{net}}(\bar{g}_B)$, and reports Aggregate-Supply CE frontier strength and tilt-recovery ratios for the Social outcome category. The baseline uses the total-assets-weighted value of $\bar{g}_B$. Direct recovery and Net-outcome recovery are signed equal-CE recovery ratios reported as fractions. Direct-tilt recovery compares the direct tilt with the Impact Frontier, while net-outcome recovery compares the unadjusted net-outcome tilt with the same frontier.

OA.3.B Frontier-strength decomposition

This subsection decomposes frontier strength into S_g, G_{dist}, C, B, R, and ϵ. Table 3 uses the decomposition to interpret why some observed proxies and constructed distributions have stronger Impact Frontiers than others.

Fix an active opportunity set $\mathcal{A}$ and let $\boldsymbol{\Sigma}_{\mathcal{A}}^{\text{eff}}$ denote the effective covariance matrix for active portfolio-weight tilts in that set. Embed the inverse of $\boldsymbol{\Sigma}_{\mathcal{A}}^{\text{eff}}$ in the full asset space as

$$\boldsymbol{\Xi}_{\mathcal{A}} = \boldsymbol{I}_{\mathcal{A}}' \left(\boldsymbol{\Sigma}_{\mathcal{A}}^{\text{eff}}\right)^{-1} \boldsymbol{I}_{\mathcal{A}}.$$

For a net outcome intensity $\boldsymbol{g}_{\text{net}}$, use the impact-return vector defined in Section 3, $\boldsymbol{r}^I = \text{diag}(\boldsymbol{P}^*)^{-1}\boldsymbol{\mathcal{C}}'\boldsymbol{g}_{\text{net}}$. Frontier strength is $\eta_{\mathcal{A}}^2(\boldsymbol{g}_{\text{net}}) = (\boldsymbol{r}^I)'\boldsymbol{\Xi}_{\mathcal{A}}\boldsymbol{r}^I$. This statistic determines the CE-return cost of producing a target portfolio impact return. If the investor targets portfolio impact return $x > 0$, the Impact Frontier requires CE-return cost

$$\Delta r_{\mathcal{A}}^{CE}(x) = \frac{x^2}{2\eta_{\mathcal{A}}^2(\boldsymbol{g}_{\text{net}})}.$$

A larger $\eta_{\mathcal{A}}^2$ means that the capital-allocation channel produces more portfolio impact return per unit of CE-return cost.

The strongest and weakest illustrative distributions in Figure 4 are constructed by treating the net outcome intensity itself as the choice vector, holding the estimated capital-allocation system and effective covariance matrix fixed. Let

$$\boldsymbol{H}_{\mathcal{A}} \equiv \boldsymbol{\mathcal{C}} \boldsymbol{\Xi}_{\mathcal{A}} \boldsymbol{\mathcal{C}}'.$$

If the outcome-intensity distribution has already been normalized to Euclidean unit length in the relevant outcome-intensity space, the frontier-strength problem is the ordinary eigenvalue problem:

$$\max_{\boldsymbol{g}:\, \boldsymbol{g}'\boldsymbol{g}=1} \boldsymbol{g}'\boldsymbol{H}_{\mathcal{A}}\boldsymbol{g} \quad \text{or} \quad \min_{\boldsymbol{g}:\, \boldsymbol{g}'\boldsymbol{g}=1} \boldsymbol{g}'\boldsymbol{H}_{\mathcal{A}}\boldsymbol{g}.$$

The leading eigenvector defines the strongest distribution, and the smallest relevant eigenvector defines the weakest distribution. With the weighted outcome-intensity normalization used in the tables, represented by a positive-definite matrix $\boldsymbol{W}_g$, the same problem becomes the generalized eigenvalue problem

$$\boldsymbol{H}_{\mathcal{A}}\boldsymbol{g} = \zeta \boldsymbol{W}_g \boldsymbol{g}.$$

The constructed distributions are implied by the calibrated frontier-strength quadratic form under the stated normalization. They do not change the observed outcome-intensity vectors or the estimated slope matrices.

To decompose $\eta^2_{\mathcal{A}}$, write the implemented net outcome intensity as a scale times a normalized vector:

$$\boldsymbol{g}_{\text{net}} = \sigma_g \tilde{\boldsymbol{g}}, \qquad S_g = \log \sigma_g,$$

where $\tilde{\boldsymbol{g}}$ has total-assets-weighted mean zero and total-assets-weighted standard deviation one. Then

$$\log \eta^2_{\mathcal{A}}(\boldsymbol{g}_{\text{net}}) = 2S_g + \log \eta^2_{\mathcal{A}}(\tilde{\boldsymbol{g}}).$$

The term S_g is the log standard deviation of the implemented net outcome intensity. It enters the log-frontier-strength decomposition as $2S_g$. The remaining terms describe whether the outcome distribution aligns with high-frontier-strength modes in the capital-allocation system.

Let the normalized-outcome frontier strength have modal contributions indexed by k:

$$\eta_{\mathcal{A}}^2(\tilde{\boldsymbol{g}}) = \sum_k \exp(a_{\mathcal{A}k}).$$

For each mode, write the log contribution as

$$a_{\mathcal{A}k} = 2G_{\mathrm{dist},\mathcal{A}k} + 2C_{\mathcal{A}k} - R_{\mathcal{A}k} + \epsilon_{\mathcal{A}k}.$$

The term $G_{\mathrm{dist},\mathcal{A}k}$ captures the normalized vector's loading on outcome dispersion in mode k. The term $C_{\mathcal{A}k}$ captures contribution strength: whether capital in that mode changes issuer scale through the contribution multiplier matrix. The term $R_{\mathcal{A}k}$ captures CE-return cost. The residual $\epsilon_{\mathcal{A}k}$ records the approximation error from the modal factorization.

Define the modal contribution shares

$$\pi_{\mathcal{A}k} = \frac{\exp(a_{\mathcal{A}k})}{\sum_j \exp(a_{\mathcal{A}j})}.$$

The log-sum identity implies

$$\log \eta_{\mathcal{A}}^2(\boldsymbol{g}_{\mathrm{net}}) = 2S_g + 2G_{\mathrm{dist},\mathcal{A}} + 2C_{\mathcal{A}} + B_{\mathcal{A}} - R_{\mathcal{A}} + \epsilon_{\mathcal{A}},$$

where

$$\begin{aligned} 2G_{\mathrm{dist},\mathcal{A}} &= \sum_k \pi_{\mathcal{A}k} 2G_{\mathrm{dist},\mathcal{A}k}, & 2C_{\mathcal{A}} &= \sum_k \pi_{\mathcal{A}k} 2C_{\mathcal{A}k}, \\ R_{\mathcal{A}} &= \sum_k \pi_{\mathcal{A}k} R_{\mathcal{A}k}, & \epsilon_{\mathcal{A}} &= \sum_k \pi_{\mathcal{A}k} \epsilon_{\mathcal{A}k}, \\ B_{\mathcal{A}} &= -\sum_k \pi_{\mathcal{A}k} \log \pi_{\mathcal{A}k}. \end{aligned}$$

The term $B_{\mathcal{A}}$ is modal breadth. It is larger when frontier strength comes from several modes rather than one dominant mode. The term $-R_{\mathcal{A}}$ is larger when the normalized vector loads on modes with lower CE-return cost.

For the rows reported in Table 3, the residual term $\epsilon_{\mathcal{A}}$ ranges from -0.13 to -0.0028, with maximum absolute residual 0.13.

OA.4 Data construction details

This section defines the sample screens and variables used in the empirical analysis.

OA.4.A Sample construction

Similar to Betermier et al. (2025), the sample uses merged CRSP-Compustat issuer data, one main listed security, return-based cost and risk variables, accounting-capital variables, and side-specific variables. Table OA.2 reports the resulting sample counts at each screen.

Sample stage	Rows	Distinct issuers
Raw CRSP–Compustat linked panel in selected years	12,419	5,858
Lagged total-assets percentile ranking universe	10,411	4,791
Expanded no-score Environmental-specification diagnostic	7,051	3,051
Final forward 3SLS estimation sample requiring the Environmental score	4,120	1,910
Pre-period risk-variable timing sample	6,346	2,678
2019 frontier and covariance cross-section	1,200	1,200

Table OA.2: **Sample construction.** This table reports sample construction for the baseline specification. The estimation sample requiring the Environmental score pools the 2004, 2009, 2014, and 2019 December 31 cross-sections at a five-year forward horizon after requiring the variables used in the simultaneous system. The pre-period risk-variable timing sample shows the coverage available for specifications that do not require the Environmental score.

OA.4.B Variable selection and construction

Environmental estimation variable in the baseline system. The baseline simultaneous system includes an MSCI Environmental score variable as an ESG-related estimation variable. This use is incidental to estimating the investor-supply and issuer-demand slopes; it does not make Environmental the preferred outcome direction for the frontier calculations. The MSCI scores are used downstream as outcome-intensity proxies in the frontier calculations rather than as additional supply or demand variables. This keeps the estimated slope system parsimonious. For each panel date, the score variables use the first available matched score observation in the window from January of the panel year through December five years later. No earlier score is substituted for missing panel-year coverage. This timing choice mainly affects early panels with sparse score coverage; the expanded no-score diagnostic reports the larger sample available when the score requirement is removed.

Lagged total-assets percentile. The empirical non-market-risk term uses a predetermined neglectedness proxy. For panel date t and horizon h_t, the ranking variable is horizon-lagged total assets, $TA_{n,t-h_t}$. The ranking sample consists of issuers with an active CRSP-Compustat link, positive market capitalization, and finite positive $TA_{n,t-h_t}$. It does not require MSCI coverage, profitability, positive current book equity, book-equity growth, complete 3SLS variables, or complete frontier covariance history.

Let N_t^{TA} be the number of issuers in the ranking sample. For $N_t^{TA} > 1$, the lagged total-assets percentile is

$$p_{n,t}^{TA} = \frac{\mathrm{rank}_t^{\mathrm{avg}}(\log TA_{n,t-h_t}) - 1}{N_t^{TA} - 1}, \tag{OA.28}$$

where $\mathrm{rank}_t^{\mathrm{avg}}$ uses the average rank for ties. If the ranking sample contains a single issuer, the percentile is set to one. The reverse size percentile $1 - p_{n,t}^{TA}$ is the neglectedness proxy used in the non-market-risk term; as in the main text, it is not an estimate of the theoretical $q_n^{-1} - 1$ term.

Business-equity correlation. The baseline business-equity-correlation variable is the correlation between issuer quarterly returns and aggregate private-business-equity growth over the forward horizon. For the 60-month baseline, the window runs from the first quarter after the panel date through the final quarter of the five-year horizon. The minimum required number of matched quarters equals the horizon length in quarters. The pre-t timing diagnostic replaces this variable with a pre-panel analogue.

Cash-flow volatility. The cash-flow-volatility control is the standard deviation of quarterly return on assets over the five-year window ending at the panel date. Quarterly return on assets is net income divided by quarterly assets.

Book equity. Book equity follows the standard Compustat construction and is set to missing when nonpositive. It enters the Appendix as an alternative demand quantity.

Profitability. Profitability is measured from the latest annual Compustat statement with statement date weakly before the panel date. The profitability variable is the Compustat operating-profitability ratio using book equity in the denominator. It requires positive book equity and nonmissing sales. There is no additional requirement that the issuer have several prior years of profitability observations. Specifications that include profitability require the ratio to be finite; specifications that exclude profitability do not impose that filter.

OA.5 Estimated supply and demand system

This section extends Section 5. It reports additional equation-by-equation estimates, identification diagnostics, non-market-risk robustness, and timing, frequency, and horizon diagnostics for the baseline supply-and-demand system. Frontier-specific alternative estimations and multiplier-scale diagnostics are reported separately in Section OA.6. Appendix regression tables use unclustered standard errors for the displayed estimator unless a table states otherwise. When parentheses report t-statistics, those statistics are computed from the same standard errors.

External elasticity translations. Table OA.3 presents more detail on the elasticity estimates in Section 5. The market-portfolio slope reported is for the baseline 2019 cross-section. For the rows from this paper, the table uses the local μ-V slope matrices and reports the reciprocal of the relevant price-quantity slope to match the relative price-elasticity magnitude reported in the literature. The issuer-side rows for this paper use diagonal demand slopes; they do not allow for cross-issuer demand-network effects.

OLS and 2SLS estimates. Table OA.4 reports the equation-by-equation OLS and 2SLS estimates for the same estimation sample as the main 3SLS system. OLS shows the raw

Source	Reported quantity	Elasticity estimate
This paper, investor side	Market-basket μ-V supply slope $m_S = V'\Phi_S^V(V/V_M) = 0.50$	$1/m_S = 2.00$
Gabaix and Koijen (2021)	Aggregate market multiplier of about 5 dollars per dollar	$1/5 = 0.20$
Koijen and Yogo (2019)	A 10 percent aggregate demand shock moves the 2017 median stock price by 26 percent	$1/(0.26/0.10) = 0.38$
Haddad et al. (2025)	Average stock-level aggregate elasticity about 0.4; passive-investing counterfactual raises the price multiplier from 2.5 to 2.9	About 0.34–0.40
Cassella et al. (2026), Table 8	Residual-demand regressions using stock performance over the 60 months after the trust-law reform	0.10–0.36
Cassella et al. (2026), Table A.2	Same residual-demand design using a 12-month return horizon	0.15–2.03
Davis et al. (2025), classical case	Classical stock-level portfolio-choice case before empirical price-impact and spanning adjustments	About 7,000, plotted as $> 5{,}000$
Davis et al. (2025), adjusted decomposition	Same stock-level portfolio-choice decomposition after empirical price-impact and unspanned-return adjustments	About 12
This paper, issuer side	Diagonal issuer-demand slope $m_{D,n} = V_n\Phi_{D,nn}^V = \lvert\Delta_D\rvert\sigma_n$; market-basket slope $V'\Phi_D^V(V/V_M) = 0.134$	Median issuer 7.28; market basket 7.44
This paper, issuer side, 12-month diagnostic	Same demand-elasticity translation scaled by the ratio of the 60- to 12-month demand coefficients in Table OA.17, 0.256/0.625	Median issuer 2.98; market basket 3.05
Choi et al. (2025)	Quarterly dynamic model with investor-demand loading $\beta_{me} = 0.3389$ and production curvature $\alpha = 0.6210$	Stock demand: $1 - \beta_{me} = 0.66$; firm demand: $1/(1-\alpha) = 2.64$

Table OA.3: **External elasticity translations.** The table reports positive elasticity magnitudes used for the scale comparisons in Section 5. For external price-impact multipliers m, the reported elasticity is $1/m$. For the rows from this paper, the reported elasticity is the reciprocal of the corresponding local μ-V price-quantity slope. The issuer-side rows from this paper use diagonal demand slopes. The rows are scale comparisons, not identical estimands; the reported quantities include aggregate market-flow, stock-level, and relative-demand translations. The 12-month row is a coefficient-scaling diagnostic. For Cassella et al. (2026), the 60-month row reflects gradual trust-law rebalancing over the event window, not a one-time shock held fixed for 60 months. For Choi et al. (2025), the firm-side number is a local structural elasticity in a quarterly dynamic model rather than a 12- or 60-month event-window response.

equation-by-equation associations, 2SLS shows the IV correction equation by equation, and the bottom block reports the Sargan and Wu–Hausman tests.

	OLS		2SLS	
	Supply	Demand	Supply	Demand
Investor Supply				
Market correlation (ρ_M)	0.49***		1.3***	
	(0.044)		(0.11)	
Non-market risk term	-96**		710***	
	(41)		(190)	
Environmental score	0.0077**		0.0011	
	(0.0035)		(0.0039)	
Cash flow volatility	-0.043		0.00058	
	(0.033)		(0.036)	
Business-equity correlation	0.077**		0.21***	
	(0.037)		(0.042)	
Issuer Demand				
Log total-assets growth, $t-5$ to t		-0.060***		-0.23***
		(0.0094)		(0.021)
Sharpe ratio (lagged)		0.13***		0.16***
		(0.015)		(0.016)
Number of observations	4120	4120	4120	4120
2SLS Diagnostics				
Exogenous instruments (Sargan)			0.85	4.5
Exogenous OLS errors (Wu-Hausman)			56***	97***

Table OA.4: **OLS and 2SLS estimates of the supply and demand system.** This table reports equation-by-equation OLS and 2SLS estimates for the same baseline estimation sample used in Table 2. The bottom block reports Sargan overidentification statistics and Wu–Hausman exogeneity statistics for the 2SLS equations. Because the investor equation contains two endogenous risk terms, the detailed first-stage F statistics are reported in Table OA.7. Stars denote statistical significance at the 0.10, 0.05, and 0.01 levels.

Issuer-clustered standard errors. For column 1 of Table 2, issuer-clustered HC1 standard errors for the market-risk, non-market-risk, and issuer-demand slope coefficients are 0.10, 210, and 0.021, compared with unclustered standard errors of 0.11, 190, and 0.021. The corresponding clustered t-statistics are 13, 3.4, and -12.

S&P 500 control. For each firm, the added variable is the difference between its return correlation with the S&P 500 and its return correlation with the market portfolio, computed over the same return window as the market-correlation variable. Table OA.5 reports the baseline system with this additional control. The market-correlation coefficient rises from 1.4 to 1.7 and remains strongly estimated. The non-market-risk coefficient rises from 700 to 1700 and remains strongly estimated. The added S&P 500-minus-market term is -8.5 and strongly estimated, consistent with incremental S&P 500 association lowering required returns.

	Baseline	+ S&P-minus-market
Investor Supply		
Market correlation (ρ_M)	1.4***	1.7***
	(0.11)	(0.18)
S&P 500-minus-market correlation		-8.5***
		(1.1)
Non-market risk term	700***	1700***
	(190)	(270)
Issuer Demand		
Log total-assets growth, $t-5$ to t	-0.24***	-0.24***
	(0.021)	(0.021)
Sharpe ratio (lagged)	0.16***	0.16***
	(0.016)	(0.016)
Diagnostics		
Minimum supply first-stage F	77	10
Supply Sargan p-value	0.33	0.38
Correlation with ρ_M	-	0.82
Number of observations	4170	4131

Table OA.5: **S&P 500 supply-control diagnostic.** This table reports the baseline system after adding $\rho_{n,t}^{SP500} - \rho_{n,t}^{M}$ to the investor-side equation, where $\rho_{n,t}^{SP500}$ is firm n's return correlation with the S&P 500, measured over the same return window. All other controls, instruments, and fixed effects follow the baseline specification.

Estimator stability. Table OA.6 compares the baseline and restricted market-risk-only estimates under 2SLS and 3SLS. The baseline coefficients and direct contribution multipliers are nearly identical across the two estimators. The restricted market-risk-only estimates differ more, so the baseline result is not materially driven by the additional 3SLS system weighting.

Panel A. Coefficients

Specification	Estimator	Market-risk coefficient (t)	NMR coefficient (t)	Issuer-demand coefficient (t)
Baseline full model	3SLS	1.4 (13)	700 (3.7)	-0.24 (-12)
Baseline full model	2SLS	1.4 (13)	690 (3.6)	-0.24 (-12)
Restricted-instrument market-risk-only	3SLS	2.4 (6.2)	-	-0.11 (-7.2)
Restricted-instrument market-risk-only	2SLS	2.6 (6.5)	-	-0.24 (-12)

Panel B. Multiplier summaries

Specification	Estimator	EW median $\mathcal{C}^{\mathrm{dir}}$	VW mean $\mathcal{C}^{\mathrm{dir}}$	Overidentification p
Baseline full model	3SLS	0.28	0.65	0.33
Baseline full model	2SLS	0.28	0.65	0.33
Restricted-instrument market-risk-only	3SLS	0.010	0.42	0.53
Restricted-instrument market-risk-only	2SLS	0.0054	0.31	0.53

Table OA.6: **Estimator stability of direct contribution multipliers.** The table compares 2SLS and 3SLS coefficient estimates and contribution-multiplier statistics for the baseline and restricted market-risk-only specifications.

First-stage and overidentification diagnostics. Table OA.7 reports first-stage and overidentification tests for the baseline simultaneous system. It also stress-tests the lagged total-assets percentile: the no-percentile-IV column removes it, while the trading-coverage-IV column replaces it with pre-panel daily trading-coverage percentile. The trading-coverage proxy is only a coverage diagnostic, not a preferred market-microstructure measure, and its demand first-stage F statistic is much closer to the no-percentile-IV column than the baseline.

	Baseline	+ tangible IV	No lagged Sharpe	Trading-coverage IV	No percentile IV
Investor-side first stages					
Market correlation (ρ_M), F	310	230	430	310	310
Non-market risk term, F	77	72	5.5	77	77
Joint Cragg–Donald weak-ID F	77	72	0.45	77	77
Demand-side first stage					
Log total assets growth, F	380	380	360	6.0	3.0
Overidentification diagnostics					
Supply equation Sargan p-value	0.33	0.27	-	0.33	0.33
Demand equation Sargan p-value	0.53	0.47	0.48	0.87	0.63
Sample					
Number of observations	4170	4162	4170	4170	4170

Table OA.7: **Identification tests for the baseline specification.** The table reports first-stage F statistics and Sargan overidentification p-values from direct alternative specifications. First-stage rows report IV weak-instrument F statistics for the named endogenous term; the joint row reports the homoskedastic Cragg–Donald statistic. For the baseline supply equation, the Stock–Yogo benchmark is the 10% maximal IV-size critical value for two endogenous regressors and three excluded instruments. The diagnostics report the instrument variation behind the system estimates. The tangible-intensity column adds a richer supply-side instrument, while the no-lagged-Sharpe and no-percentile-IV columns remove the two load-bearing first-stage anchors. The trading-coverage-IV column replaces the lagged total-assets percentile with the pre-panel daily trading-coverage percentile, a diagnostic proxy defined as the fraction of benchmark-valid trading days in the prior 36 months with valid CRSP daily total-return observations, ranked from 0 to 1 within panel year. Sargan rows report p-values where overidentification is available; – denotes exact identification or unavailable tests.

The baseline first-stage F statistics are 310 for market correlation, 77 for the non-market-risk term, and 380 for total-assets growth, while the supply and demand Sargan p-values are 0.33 and 0.53. For the supply equation's two endogenous risk terms and three excluded supply instruments, the baseline joint Cragg–Donald weak-identification statistic is 77, above the Stock–Yogo 10 percent maximal IV-size critical value of 13 for the same $K_1 = 2, L_1 = 3$ design (Stock and Yogo 2005). The first-stage failures are informative: removing lagged Sharpe lowers the non-market-risk first-stage F statistic to 5.5, and removing or proxying the lagged total-assets percentile lowers the demand first-stage F statistic to 3.0 or 6.0. Together, these diagnostics show that first-stage relevance comes from distinct excluded instruments on the two sides of the system. They do not establish the exclusion restrictions, which remain identifying assumptions.

Demand-quantity diagnostics. The baseline issuer-demand quantity is log total-assets growth. Table OA.8 compares the baseline log total-assets-growth specification with alternative estimated issuer-demand quantities and fit diagnostics. Nonzero Box-Cox growth-ratio diagnostics have weaker fit or weaker first-stage strength than the log specification. The total-assets growth ratio $TA_{n,t}/TA_{n,t-5}$ keeps the negative demand sign but fits materially worse. The no-scale total-assets-change specification has the wrong demand sign. Book-equity growth is an alternative accounting-capital specification. The matched-sample rows preserve the negative issuer-demand relation and similar direct contribution multipliers.

Demand quantity or diagnostic	N	Coefficient	t	First-stage F	Sargan p	Issuer SSR / baseline	Worst-decile SSR / baseline
Log total-assets growth, baseline	4170	-0.24	-12	380	0.53	1.0	1.0
Log total-assets growth, book-equity matched sample	3944	-0.34	-11	220	0.54	1.1	-
Log book-equity growth, matched sample	3944	-0.28	-11	170	0.87	1.2	-
Box-Cox growth ratio, alpha -0.050	4170	-0.26	-12	380	0.51	1.0	1.0
Midpoint growth	4170	-0.37	-12	320	0.42	1.0	1.1
Total-assets growth ratio $TA_{n,t}/TA_{n,t-5}$	4170	-0.013	-7.5	29	0.91	2.5	11
No-scale total-assets change	4170	3.5×10^{-5}	8.3	42	0.54	2.0	12
Forward log total-assets growth, matched sample	2223	-0.25	-4.7	120	< 0.001	-	-
Leave-one-panel-out, weakest t-statistic	2833	-0.16	-7.3	280	0.079	-	-

Table OA.8: **Demand-quantity diagnostics.** The table compares the baseline log total-assets-growth specification with alternative estimated issuer-demand quantities. Issuer SSR is the issuer-equation residual sum of squares normalized by the baseline row. Worst-decile SSR reports the largest lagged-total-assets-decile issuer SSR ratio, also normalized by the baseline. The book-equity rows use the matched positive-book-equity sample and express the fitted slope in the same market-value units used for investor supply. Table OA.9 reports the corresponding multiplier and frontier sensitivity summaries for valid demand calibrations.

All reported rows retain a negative issuer-demand coefficient. Direct-tilt recovery remains close to one, while the multiplier level and net-outcome recovery are more sensitive to the demand calibration. The exercise supports the sign of the local demand relation within the

Demand calibration	Row type	Median C^{dir}	η^2 range	Direct recovery	Net-outcome recovery
Baseline log TAG	estimated	0.28	2.4–9.6	99.6–99.9	35.7–59.7
Best nonzero Box-Cox, alpha -0.050	estimated	0.27	2.2–9.1	99.6–99.9	35.0–59.0
Total-assets growth ratio	estimated	0.82	84–120	99.9–100.0	84.3–91.0
Leave-one-panel-out, drop 2009	estimated	0.36	3.9–14	99.6–99.9	38.8–63.4
Midpoint total-assets growth	estimated	0.15	1.1–4.1	99.4–99.8	40.4–61.3
Winsorized log TAG 1/99	estimated	0.26	2.1–8.6	99.6–99.9	35.0–58.8
Winsorized log TAG 5/95	estimated	0.22	1.5–6.5	99.6–99.9	33.5–56.8
Log book-equity growth	matched	0.25	2.0–8.1	99.6–99.9	34.7–58.5
Log TAG, positive-book-equity sample	matched	0.22	1.5–6.4	99.7–99.9	33.4–56.8
Midpoint book-equity growth	matched	0.16	1.2–5.2	99.0–99.9	36.5–55.3
Forward log total-assets growth	matched	0.27	2.3–9.2	99.6–99.9	35.4–59.3
Lagged log TAG, forward-feasible sample	matched	0.30	2.7–11	99.6–99.9	36.4–60.6
Lagged denominator calibration	scale check	0.17	28–130	98.8–99.9	91.3–98.2

Table OA.9: **Issuer-demand calibration and frontier sensitivity.** The table recomputes the full cross-section frontier using issuer-demand specifications with a negative coefficient and positive implied demand slopes. Estimated rows re-estimate the issuer-demand equation after changing the demand quantity or estimation window. Matched rows re-estimate the equation on the smaller samples needed for the book-equity and forward-growth comparisons. The scale-check row keeps the baseline coefficient and changes only the denominator used to translate that coefficient into demand slopes. Each row reports one valid issuer-demand calculation; frontier columns are ranges across the Social, Environmental, Governance, and Carbon Emissions outcome directions. Direct recovery and Net-outcome recovery are signed equal-CE recovery percentages. The baseline supply matrix and Aggregate-Supply CE specification are held fixed. Rows with the wrong demand sign or nonpositive implied demand slopes are documented in Table OA.8 and excluded from this frontier table.

reported specification family; it does not turn the baseline coefficient into a specification-invariant structural parameter.

OA.5.A Non-market-risk robustness

Direct contribution multiplier sensitivity to investor-supply specification. Table OA.10 separates three distinct market-risk-only exercises. The market-risk-block row removes the non-market-risk block without re-estimation and is a decomposition of the baseline. The same-instrument market-risk-only specification fails its overidentification test. The restricted market-risk-only specification removes the investor-side instrument whose rationale is specific to the non-market-risk channel; it retains a strong first stage and does not fail the reported overidentification test. Its equal-weight median direct contribution multiplier is much smaller than the baseline, while its value-weighted mean remains material. The baseline typical-issuer multiplier is conditional on the structured non-market-risk supply term.

The typical-issuer multiplier is highly sensitive to the structured non-market-risk term, but the portfolio-rule comparison is less fragile. Across the market-risk-only rows, direct-tilt recovery remains high, while net-outcome recovery falls materially below the baseline. Frontier strength does not move monotonically because the alternative estimations change both investor-supply and issuer-demand slopes.

Neglectedness-proxy diagnostics. Table OA.12 collects diagnostics for the lagged total-assets percentile and its role in the non-market-risk term. Together, the panels separate demand-side scale relevance from investor-side product/component and stability tests. In Panel A, replacing the lagged total-assets percentile with log lagged total assets preserves the negative issuer-demand coefficient and first-stage strength, so the demand-slope sign is not specific to the percentile transformation.

Anomaly and neglectedness-proxy diagnostics. A natural concern is that the non-market-risk term may capture familiar cross-sectional return premia rather than a supply-side cost of residual risk. Table OA.13 presents alternative estimations which augment the

Panel A. Direct contribution multipliers

Specification	Status	EW median $\mathcal{C}^{\text{dir}}$	VW mean $\mathcal{C}^{\text{dir}}$	TAW mean $\mathcal{C}^{\text{dir}}$
Baseline full model	Passes reported diagnostics	0.28	0.65	0.47
Market-risk block only, no re-estimation	Decomposition	0.0029	0.23	0.15
Same-instrument market-risk-only	Rejected by overidentification test	0.0027	0.22	0.14
Restricted-instrument market-risk-only	Passes reported diagnostics	0.010	0.42	0.31
Market-risk-only, NMR instruments removed from both equations	Weak demand first stage	0.0025	0.21	0.14
Minimal market-risk-only	Wrong demand sign	-	-	-

Panel B. First-stage and overidentification diagnostics

Specification	Status	Minimum first-stage F	Overidentification p
Baseline full model	Passes reported diagnostics	77	0.33
Market-risk block only, no re-estimation	Decomposition	-	-
Same-instrument market-risk-only	Rejected by overidentification test	310	< 0.001
Restricted-instrument market-risk-only	Passes reported diagnostics	38	0.53
Market-risk-only, NMR instruments removed from both equations	Weak demand first stage	3.0	0.54
Minimal market-risk-only	Wrong demand sign	0.051	-

Table OA.10: **Investor-supply specification and market-risk-only direct contribution multipliers.** The table reports contribution-multiplier statistics and identification diagnostics for the baseline, a market-risk-block decomposition, and alternative market-risk-only specifications. EW median is the equal-weight median, VW mean is the market-value-weighted mean, and TAW mean is the total-assets-weighted mean. The market-risk-block row sets the non-market-risk supply block to zero without re-estimation. Rows marked as rejected, weakly identified, or wrong-sign are diagnostics rather than alternative baseline estimates.

Supply specification	Outcome	η^2	Portfolio impact return at 100 bps	Direct recovery	Net-outcome recovery
Baseline full model	Social	4.0	0.28	99.7	47.7
Baseline full model	Environmental	9.6	0.44	99.8	35.7
Baseline full model	Governance	2.4	0.22	99.9	41.2
Baseline full model	Carbon Emissions	7.3	0.38	99.6	59.7
Market-risk block only, no re-estimation	Social	1.5	0.17	98.8	20.6
Market-risk block only, no re-estimation	Environmental	4.1	0.29	99.1	14.4
Market-risk block only, no re-estimation	Governance	0.83	0.13	99.2	20.2
Market-risk block only, no re-estimation	Carbon Emissions	2.6	0.23	98.0	15.7
Restricted-instrument market-risk-only	Social	6.3	0.36	98.8	25.2
Restricted-instrument market-risk-only	Environmental	15.0	0.55	98.7	18.3
Restricted-instrument market-risk-only	Governance	3.6	0.27	99.1	25.5
Restricted-instrument market-risk-only	Carbon Emissions	11.0	0.46	97.5	19.5

Table OA.11: **Investor-supply specification and frontier sensitivity.** The table recomputes the full cross-section frontier under selected investor-supply specifications. Rows use the baseline full cross-section and Aggregate-Supply CE for the relevant investor-supply specification. Identification status and contribution-multiplier statistics for these and rejected market-risk-only specifications are reported in Table OA.10. Portfolio impact return at 100 bps is the model-implied expected outcome change per unit investor wealth at a CE-return cost of 100 basis points. Direct recovery and Net-outcome recovery are signed equal-CE recovery percentages relative to the Impact Frontier at the same CE-return cost. The baseline, market-risk-block, and restricted market-risk-only rows are reported as calibration sensitivities. The additive-neglectedness stress test remains a coefficient-level exercise because it does not identify a separate supply matrix for frontier calculations.

investor-supply equation with controls designed to absorb common alternative interpretations. The table reports representative alternatives: a standalone additive neglectedness-proxy term $1-p^{TA}_{n,t}$, forward idiosyncratic volatility, log-lagged market capitalization, Fama–French factor betas estimated from pre-panel monthly returns, and profitability and investment characteristics (Fama and French 2015; Hou et al. 2015).

The non-market-risk coefficient remains positive, statistically significant, and close to the baseline in the idiosyncratic-volatility, size, factor-beta, and profitability-investment rows. The additive-neglectedness row differs: when the standalone neglectedness-proxy term is added to the baseline, the non-market-risk coefficient falls to about 0.25 of the baseline and both the market-risk and non-market-risk estimates become imprecise. This is not because $1-p^{TA}_{n,t}$ is mechanically the same variable as the full non-market-risk product: its Pearson correlation with that product is only 0.03. Rather, the neglectedness proxy is moderately correlated with ρ_M (-0.43) and with the residual-correlation and idiosyncratic-share components of the non-market-risk term (-0.33 and 0.43). The table supports a specific interpretation: the coefficient is robust to standard anomaly controls and the data contain a neglectedness-related investor-supply signal, but they do not separately pin down

Panel A. Alternative scale information in the demand first stage

Scale instrument	$\hat{\Delta}_D$	t	Demand F	Sargan p
Baseline percentile	-0.24	-12	380	0.53
Log lagged total assets	-0.20	-11	430	0.51
Lagged total-assets deciles	-0.13	-8.2	120	< 0.001
No scale instrument	-0.49	-1.6	3.0	0.63

Panel B. Non-market-risk product and additive component controls

Specification	Product	t	Neglect.	t	ρ_{int}	t	Fit corr.	Supply F
Baseline full product	700	3.7	-	-	-	-	1.0	310
Baseline + residual correlation	720	3.6	-	-	-7.7	-2.4	1.0	290
ρ_M + neglectedness proxy + residual correlation	-	-	0.47	3.1	5.7	2.4	0.87	30
ρ_M + neglectedness proxy × residual correlation	510	3.7	-	-	-	-	0.99	310

Panel C. Stability and neglectedness-tail influence

Diagnostic	Coef. or range	t or range	Supply F or range	Sargan p or range
Leave-one-panel-out baseline supply	550–1100	2.9–4.5	200–290	-
Leave-one-panel-out log lagged-TA IV	570–1100	2.9–4.7	180–250	0.30–0.62
Drop least-neglected decile	710	3.7	280	0.19
Drop most-neglected decile	760	3.6	160	0.60
Drop both extreme neglectedness deciles	840	4.0	150	0.44
Winsorize neglectedness proxy at panel p10/p90	750	3.9	300	0.29

Table OA.12: **Neglectedness-proxy diagnostics for the non-market-risk term.** Panel A presents alternative estimations of the issuer-demand equation with alternative scale instruments for log total-assets growth. Panel B presents alternative estimations of the investor-supply equation to compare the structured non-market-risk product with residual-correlation and component controls. Panel C reports leave-one-panel-out and neglectedness-tail tests for the non-market-risk coefficient. The neglectedness proxy is $1 - p_{n,t}^{TA}$, where $p_{n,t}^{TA}$ is the lagged total-assets percentile. The additive component rows test whether the data distinguish the structured product from unrestricted functions of its own components. The baseline-plus-standalone-neglectedness specification is reported in Table OA.13, which reports the coefficient-level stress test with anomaly controls and standard errors. Fitted-risk correlation is the cross-sectional correlation with the baseline fitted investor-side risk term.

Panel A. Coefficients and added controls

Specification	ρ_M (s.e.)	Non-market (s.e.)	Neglect. (s.e.)	NM/base	Controls
Baseline	$1.4***$ (0.11)	$700***$ (190)	-	1.0	-
+ neglect.	2.6 (4.2)	180 (1900)	0.36 (1.2)	0.25	$1 - p_{TA}$
+ idio vol	$1.6***$ (0.14)	$870***$ (160)	-	1.2	Idio vol.
+ size	$3.0***$ (0.58)	$1200***$ (390)	-	1.7	Lagged market cap
+ FF factor betas	$1.3***$ (0.12)	$760***$ (210)	-	1.1	FF5 non-market betas
+ prof./inv.	$1.3***$ (0.11)	$800***$ (220)	-	1.1	Profitability; invest.

Panel B. First-stage and overidentification diagnostics

Specification	F_{ρ_M}	Non-market F	Wu p	Sargan p	N
Baseline	310	77	< 0.001	0.33	4170
+ neglect.	28	110	< 0.001	-	4170
+ idio vol	140	88	< 0.001	0.61	4170
+ size	110	250	< 0.001	0.62	4170
+ FF factor betas	240	67	< 0.001	0.35	4138
+ prof./inv.	440	95	< 0.001	-	4168

Table OA.13: **Anomaly and neglectedness-proxy diagnostics for the investor-supply equation.** The table reports alternative estimations of the baseline simultaneous system after adding the listed controls to the investor-supply equation. Coefficient rows report 3SLS estimates with standard errors in parentheses. The NM/base column divides the non-market-risk coefficient by the baseline estimate. The neglectedness column reports the coefficient on standalone $1 - p_{n,t}^{TA}$ when it is included. Panel A reports coefficient estimates and added controls. Panel B reports first-stage and overidentification tests. First-stage and overidentification diagnostics are from the investor-supply 2SLS equation. The factor-beta row adds Fama–French betas estimated from pre-panel monthly returns. Stars denote statistical significance at the 0.1, 0.05, and 0.01 levels.

the structured multiplicative channel and a free additive neglectedness effect.

Neglectedness-proxy functional-form robustness. The theory motivates a neglectedness factor related to $q_n^{-1} - 1$, while the empirical baseline uses the parsimonious linear proxy $1 - p_{n,t}^{TA}$. Table OA.14 replaces that proxy with standard monotone alternatives. The diagnostic tests whether the non-market-risk estimate depends on this functional-form choice.

Proxy form	Proxy factor	Coef.	SE	Median $\mathcal{C}^{\text{dir}}$	Social net-outcome recovery
Linear reverse size percentile	$(1-p)$	700	190	0.28	0.48
Quadratic reverse size percentile	$(1-p)^2$	1400	390	0.19	0.48
Cubic reverse size percentile	$(1-p)^3$	2200	640	0.13	0.48
Floored hyperbolic	$((p^{\dagger})^{-1} - 1)$	180	56	0.15	0.46

Table OA.14: **Non-market-risk neglectedness-proxy functional-form robustness.** The table reports robustness diagnostics for the neglectedness proxy used in the non-market-risk term. Coefficient and standard-error columns report the 3SLS coefficient on the resulting non-market-risk term. Median $\mathcal{C}^{\text{dir}}$ is the median direct contribution multiplier in the 2019 frontier cross-section. The final column reports equal-CE net-outcome recovery for the Social outcome category.

The alternative forms keep the non-market-risk coefficient positive and precisely estimated. The median direct contribution multiplier is lower under the convex and floored-hyperbolic forms, while the net-outcome recovery ratio remains close to the baseline. Thus the qualitative multiplier result and the portfolio comparison do not depend on the linear form, although the calibrated multiplier level is functional-form sensitive. The paper retains the linear reverse-percentile form because it is transparent and avoids making the smallest issuers carry disproportionate weight.

OA.5.B Timing, frequency, and horizon diagnostics

Forward demand. The default issuer-demand equation uses $\log(TA_{n,t}/TA_{n,t-5})$, because this balance-sheet quantity is observed at the panel date. The forward total-assets-growth coefficient remains negative and precisely estimated on the feasible 2004, 2009, and 2014 sample. The matched lagged and forward rows in Table OA.9 show that the timing change does not reverse the local demand calibration, although the smaller forward sample is not used as the baseline.

Pre-t risk-variable timing. The main timing concern is that the baseline investor-side equation measures the forward Sharpe-ratio proxy and the forward risk variables on the same realized return window. The pre-t risk-variable diagnostic keeps the forward Sharpe-ratio left-hand side, keeps total-assets growth as the issuer-demand quantity, and replaces the investor-side market-risk, non-market-risk, and business-equity-correlation variables with pre-panel or as-of-panel analogues. This breaks the shared return window while exploring whether expected horizon risk is priced in required returns. Table OA.15 reports the comparison.

	Forward baseline	Pre-t risk-variable diagnostic
Investor-side coefficients		
Market risk term	1.4 (13)	1.2 (8.2)
Non-market risk term	700 (3.7)	1500 (6.9)
Issuer-demand coefficient		
Log total-assets growth	-0.24 (-12)	-0.30 (-9.7)
First-stage diagnostics		
Market risk term, F	310	210
Non-market risk term, F	77	48
Log total-assets growth, F	380	210
Overidentification diagnostics		
Investor-side Sargan p-value	0.33	0.69
Issuer-demand Sargan p-value	0.53	< 0.001
Sample		
Number of observations	4170	3820

Table OA.15: **Pre-t risk-variable timing.** The table compares the forward baseline with an alternative estimation that keeps the forward Sharpe-ratio left-hand side but measures the investor-side risk variables and business-equity correlation before the panel date. The diagnostic tests whether the investor-side risk estimates depend mechanically on using the same realized return window on both sides of the supply equation. It is used to assess sign stability and statistical significance. Coefficient rows report 3SLS estimates with t-statistics in parentheses.

The alternative estimation keeps the market-risk and non-market-risk terms positive and strongly estimated when the shared forward return window is broken. The issuer-demand slope also remains negative. At the same time, it has a smaller sample and weaker issuer-demand overidentification statistics than the forward baseline. It supports the view that the non-market-risk result is not only a shared-window artifact.

Return frequency for ρ_M. The monthly baseline is a five-year capital-allocation calibration. The return-frequency diagnostic tests whether measuring ρ_M with weekly or available daily return inputs overturns the baseline 3SLS supply-and-demand estimates. Table OA.16 reports the resulting coefficient and overidentification diagnostics.

Return input	ρ_M horizon	Market-risk	Non-market coef. (t)	Issuer demand	Sargan p supply/demand
Monthly	Native monthly	1.4 (13)	700 (3.7)	-0.24 (-12)	0.33/0.53
Weekly	Native weekly	1.3 (14)	180 (0.66)	-0.22 (-12)	0.42/0.45
Weekly	4W overlap	1.4 (13)	620 (2.2)	-0.23 (-11)	0.24/0.45
Weekly	8W overlap	1.5 (12)	870 (3.1)	-0.22 (-11)	0.15/0.45
Weekly	12W overlap	1.5 (11)	1100 (3.7)	-0.22 (-11)	0.11/0.45
Daily diagnostic	Native daily	1.3 (13)	-600 (-1.3)	-0.22 (-11)	0.33/0.55
Daily diagnostic	21D overlap	1.3 (12)	-74 (-0.16)	-0.22 (-11)	0.18/0.55
Daily diagnostic	42D overlap	1.5 (11)	70 (0.14)	-0.22 (-11)	0.12/0.55
Daily diagnostic	63D overlap	1.7 (10)	130 (0.24)	-0.22 (-11)	0.12/0.55

Table OA.16: **Return-frequency and market-correlation tests.** The table reports return-frequency diagnostics for the 3SLS supply-and-demand estimates. The monthly row is the baseline. High-frequency rows replace the return input used to measure ρ_M and the associated non-market-risk term. Coefficient cells report 3SLS estimates with t-statistics in parentheses. Sargan p-values are from the corresponding 2SLS equations.

The market-risk coefficient is stable across return-frequency branches, ranging from 1.3 to 1.7 with t-statistics from 10 to 14. The non-market-risk coefficient is informative for the monthly baseline and overlapping weekly horizons, including 620 at four weeks and 1100 at twelve weeks, but it is weak in every daily branch, with negative point estimates in the native-daily and 21-day-overlap rows. This pattern is consistent with horizon sensitivity and greater exposure to short-interval noise, non-synchronous trading, and thin-trading effects. It supports interpreting the monthly baseline as a five-year capital-allocation calibration rather than a daily-trading calibration. The monthly baseline remains the specification used in the frontier calculations.

Horizon analysis. The 60-month horizon remains the baseline specification. The horizon variants diagnose whether the estimated issuer and investor slopes vary with the time window set to 12, 24, 36, 48, and 60 months. Table OA.17 reports whether the five-year baseline is better interpreted as a short-run price-pressure result or as a multi-year response to a sustained tilt.

Horizon	N	ρ_M	Non-market	Demand	$\widehat{\epsilon}_D$	Min. F	Med. C^{dir}	sd(h)	η^2	corr(h, h_{60})	Net-outcome recovery
12	24425	0.97 (8.5)	2200 (1.6)	-0.63 (-8.6)	1.6	58	0.46	0.76	2.8	0.90	0.58
24	23278	1.2 (17)	1100 (2.3)	-0.38 (-16)	2.6	180	0.25	0.42	3.1	0.95	0.48
36	21698	1.2 (21)	750 (3.4)	-0.32 (-20)	3.1	430	0.21	0.36	3.4	0.97	0.44
48	20168	1.2 (23)	370 (2.0)	-0.29 (-22)	3.5	320	0.14	0.27	3.2	0.99	0.40
60	18724	1.3 (25)	380 (2.9)	-0.26 (-24)	3.9	430	0.17	0.32	4.0	1.0	0.39

Table OA.17: **Horizon diagnostics.** The table reports horizon-specific estimates and contribution diagnostics for the baseline specification family, re-estimated on annual diagnostic panels from 2004 to 2020. The 60-month row remains the baseline specification; shorter horizons are reported as tests for the multi-year response interpretation. Coefficient cells report estimate and t-statistic from the baseline 3SLS specification. $\widehat{\epsilon}_D = -1/\widehat{\Delta}_D$. Multiplier and frontier columns use the latest common annual diagnostic panel, 2020. $\mathcal{C}_n^{\text{dir}} = \Phi_{S,nn}/(\Phi_{S,nn} + \Phi_{D,nn})$ is the direct contribution multiplier. The $sd(h)$, η^2, corr(h, h_{60}), and net-outcome recovery columns average across the Social, Environmental, Governance, and Carbon Emissions outcome categories. $\eta^2 = h'(\hat{\mathbf{\Sigma}}_{S,t}^{\text{eff}})^{-1}h$ is the Aggregate-Supply CE frontier denominator, where $\hat{\mathbf{\Sigma}}_{S,t}^{\text{eff}}$ is the Aggregate-Supply CE effective covariance matrix for the 2020 panel, and net-outcome recovery is the equal-CE portfolio impact return of the net-outcome tilt relative to the Impact Frontier, with the sign retained.

The horizon tests support the paper's multi-year interpretation while showing that the estimated response system is horizon-specific. The impact-intensity vector stabilizes relatively quickly: the correlation with the 60-month vector is 0.90 at 12 months, 0.95 at 24 months, and 0.97 at 36 months, rising to 0.99 by 48 months. The slope coefficients and median direct contribution multiplier, however, do not move by a common scale factor. The market-risk coefficient rises with the horizon, the non-market-risk coefficient generally falls, issuer demand generally becomes more elastic, and the median direct contribution multiplier is non-monotone. The one-year row is also weaker as a capital-adjustment diagnostic: its non-market coefficient has a t-statistic of 1.6, compared with 2.3, 3.4, 2.0, and 2.9 at the longer horizons. Average net-outcome tilt recovery also falls from 0.58 at 12 months to 0.39 at 60 months. The direct contribution multiplier remains interior at every reported horizon, but the five-year row is a distinct multi-year calibration rather than a less noisy estimate of a horizon-invariant system.

Demand-cluster slopes and off-diagonal issuer demand. This empirical robustness diagnostic tests whether a narrow demand-cluster term can be estimated in the issuer-demand equation and used to construct off-diagonal issuer-demand slopes. The baseline keeps $\mathbf{\Phi}_D$ diagonal; this diagnostic tests whether one selected SIC3 link specification changes the frontier comparisons. It is motivated by the possibility that issuers in the same narrow industry share financing windows, competitive pressure, or cluster-level substitution patterns that are not fully absorbed by the year and industry indicators in the baseline specification.

The baseline issuer-demand equation uses own log total-assets growth. The demand-cluster diagnostic augments the issuer equation with leave-one-out SIC3 total-assets growth, defined as the lagged-total-assets-weighted aggregate growth of the other issuers in issuer n's SIC3 group. The selected diagnostic specification uses leave-one-out SIC3 growth and applies the off-diagonal term only to SIC3 clusters with at least six usable issuers in the relevant panel year. The fitted demand-cluster coefficient is then used to construct off-diagonal entries of Φ_D: for issuer m in issuer n's leave-one-out SIC3 group, the signed demand slope is proportional to $\hat{\kappa}\,TA_m/TA_{-n,G(n)}$, with the usual volatility and μ-V scaling applied before frontier calculations. Table OA.18 reports the fitted coefficient and the resulting recovery ranges; the diagnostic changes the off-diagonal entries of $\mathbf{\Phi}_D$ and the system-response term. It is not a general demand-network estimate.

Diagnostic	Cluster coef. (t)	First-stage F	Sargan p	Baseline recovery	Cluster recovery
Panel A. Selected SIC3 demand-cluster equation					
SIC3 leave-one-out cluster growth	-0.20 (-2.0)	49	0.22	-	-
Panel B. Observed equal-CE recovery ranges					
Market, Aggregate-Supply CE, Direct	-	-	-	1.0–1.0	0.69–0.94
Market, Aggregate-Supply CE, Net-outcome	-	-	-	0.36–0.60	0.34–0.53
SIC2, Market-Risk CE, Direct	-	-	-	0.82–1.0	-0.0081–1.0
SIC2, Market-Risk CE, Net-outcome	-	-	-	0.17–1.0	0.088–0.98

Table OA.18: **SIC3 demand-cluster diagnostic and equal-CE recovery ranges.** Panel A reports the selected leave-one-out SIC3 demand-cluster coefficient, its added first-stage statistic, and the issuer-demand Sargan p-value. Panel B compares observed signed equal-CE recovery ranges before and after using that coefficient to construct off-diagonal entries of Φ_D. Market rows use the four outcome categories under Aggregate-Supply CE. SIC2 rows use observed outcome-by-SIC2 cases with at least five issuers under Market-Risk CE; constructed worst cases are excluded from these ranges. The exercise is a selected network diagnostic, not a preferred replacement for the diagonal issuer-demand baseline.

The demand-cluster extension widens the gap between the direct tilt and the Impact

Frontier: direct recovery ranges fall in the full cross-section and SIC2-restricted cases, while net-outcome recovery remains materially below the frontier and generally also falls. Direct tilts perform well in the baseline full cross-section, but richer issuer-demand interactions can make the full contribution multiplier matrix more important; the evidence here comes from one selected SIC3 construction rather than a broad search over demand networks.

OA.6 Frontier results

This section collects diagnostics tied to the frontier figures in Section 6. The organizing distinction is frontier scale versus portfolio-rule recovery: absolute frontier magnitudes are calibration-sensitive, while direct-tilt recovery remains high and net-outcome recovery remains materially lower across the reported observed full cross-section specifications. Table OA.21 reports a score-noise sensitivity check that lowers outcome-score reliability while holding the estimated supply-and-demand system fixed.

Effective-covariance sensitivity. Table OA.19 recomputes the full cross-section frontier under alternative positive-definite effective covariance matrices. Direct-tilt recovery remains close to one under every reported specification, while net-outcome recovery remains materially lower. Frontier strength is more sensitive, especially when the CE specification changes from Aggregate-Supply CE to Market-Risk CE. The equal-CE impact-return gap is not an artifact of one covariance regularization, although frontier strength changes with the effective covariance matrix.

Score-cardinality sensitivity. Table OA.20 changes only the cardinal representation of the score proxy before applying the baseline risk-free-asset netting.

Nonlinear score transformations materially change frontier units and scale, while direct-tilt recovery remains near one and net-outcome recovery remains positive but materially lower.

Score-noise sensitivity. Table OA.21 attenuates the observed outcome score toward a noisy proxy with reliability ρ_g while holding the estimated supply-and-demand system fixed.

Effective covariance	Outcome	Frontier strength η^2	Impact return at 100 bps	Direct recovery	Net-outcome recovery
Baseline regularization	Social	4.0	0.28	99.7	47.7
Baseline regularization	Environmental	9.6	0.44	99.8	35.7
Baseline regularization	Governance	2.4	0.22	99.9	41.2
Baseline regularization	Carbon Emissions	7.3	0.38	99.6	59.7
Weaker shrinkage ($\theta = 0.48$)	Social	6.0	0.35	99.8	45.7
Weaker shrinkage ($\theta = 0.48$)	Environmental	15	0.54	99.9	33.3
Weaker shrinkage ($\theta = 0.48$)	Governance	3.7	0.27	99.9	39.9
Weaker shrinkage ($\theta = 0.48$)	Carbon Emissions	10	0.45	99.7	64.4
Stronger shrinkage ($\theta = 1$)	Social	3.9	0.28	99.7	47.8
Stronger shrinkage ($\theta = 1$)	Environmental	9.4	0.43	99.8	36.0
Stronger shrinkage ($\theta = 1$)	Governance	2.3	0.21	99.9	41.3
Stronger shrinkage ($\theta = 1$)	Carbon Emissions	7.2	0.38	99.6	59.0
Diagonal residual covariance	Social	4.0	0.28	99.7	47.8
Diagonal residual covariance	Environmental	9.7	0.44	99.8	35.8
Diagonal residual covariance	Governance	2.4	0.22	99.9	41.2
Diagonal residual covariance	Carbon Emissions	7.4	0.38	99.6	59.9
Market-Risk CE	Social	100	1.4	99.8	47.6
Market-Risk CE	Environmental	240	2.2	99.9	40.6
Market-Risk CE	Governance	70	1.2	100.0	43.8
Market-Risk CE	Carbon Emissions	230	2.2	99.8	42.6

Table OA.19: **Effective-covariance sensitivity of frontier scale and recovery.** The table recomputes the full cross-section frontier under alternative positive-definite effective covariance matrices. Regularization variants are compared with the baseline Aggregate-Supply CE specification. The baseline row uses the Aggregate-Supply CE effective covariance matrix and residual-covariance shrinkage weight $\omega_t = 0.952$. Market-Risk CE is reported separately because it changes the CE specification rather than only the regularization rule. Portfolio impact return at 100 bps is the model-implied expected outcome change at a CE-return cost of 100 basis points. Direct recovery and Net-outcome recovery are signed equal-CE recovery percentages relative to the Impact Frontier at the same CE-return cost.

Outcome	Score transform	Frontier strength η^2	Impact return at 100 bps	Direct recovery	Net-outcome recovery	Corr. raw	Corr. net
Social	Baseline raw	4.0	0.28	99.7	47.7	1.00	1.00
Social	Rank normalized	0.16	0.057	99.7	43.9	0.96	0.97
Social	Z-scored raw	2.0	0.20	99.7	47.7	1.00	1.00
Social	Winsorized 5/95	3.2	0.25	99.7	45.2	0.98	0.98
Social	Rank Gaussian	2.0	0.20	99.7	47.5	1.00	1.00
Environmental	Baseline raw	9.6	0.44	99.8	35.7	1.00	1.00
Environmental	Rank normalized	0.12	0.049	99.8	36.0	0.98	0.99
Environmental	Z-scored raw	2.1	0.20	99.8	35.7	1.00	1.00
Environmental	Winsorized 5/95	7.0	0.37	99.8	34.6	0.99	1.00
Environmental	Rank Gaussian	1.8	0.19	99.8	37.0	0.99	1.00
Governance	Baseline raw	2.4	0.22	99.9	41.2	1.00	1.00
Governance	Rank normalized	0.12	0.050	99.9	38.8	0.99	0.99
Governance	Z-scored raw	1.5	0.17	99.9	41.2	1.00	1.00
Governance	Winsorized 5/95	2.1	0.20	99.9	39.7	0.99	0.99
Governance	Rank Gaussian	1.5	0.17	99.9	41.0	1.00	1.00
Carbon Emissions	Baseline raw	7.3	0.38	99.6	59.7	1.00	1.00
Carbon Emissions	Rank normalized	0.15	0.055	99.6	62.7	0.96	0.96
Carbon Emissions	Z-scored raw	1.7	0.18	99.6	59.7	1.00	1.00
Carbon Emissions	Winsorized 5/95	6.9	0.37	99.6	62.0	0.99	0.99
Carbon Emissions	Rank Gaussian	1.9	0.19	99.6	61.4	0.99	0.99

Table OA.20: **Score-cardinality sensitivity.** The table recomputes the observed score-based frontiers after transparent transformations of the same MSCI inputs. These rows are diagnostics for the cardinal representation of the outcome-intensity proxy, not alternative estimates of true marginal external effects. All transformations retain the baseline orientation and risk-free-asset netting, which is applied after each score transformation. Corr. raw and Corr. net report unweighted Pearson correlations, across issuers with finite paired values, of the transformed gross and net outcome-intensity vectors with their corresponding baseline vectors. Direct recovery and Net-outcome recovery are signed equal-CE recovery percentages for the direct tilt and net-outcome tilt.

Lower score reliability reduces frontier strength and impact return at a fixed CE-return cost but leaves equal-CE recovery unchanged in this diagnostic. The exercise changes score reliability, not the cardinal spacing of the observed score.

Reliability ρ_g	Outcome	Frontier strength	Impact return at 100 bps	Direct recovery	Net-outcome recovery
1.00	Social	4.0	0.28	99.7	47.7
1.00	Environmental	9.6	0.44	99.8	35.7
1.00	Governance	2.4	0.22	99.9	41.2
1.00	Carbon Emissions	7.3	0.38	99.6	59.7
0.75	Social	2.2	0.21	99.7	47.7
0.75	Environmental	5.4	0.33	99.8	35.7
0.75	Governance	1.3	0.16	99.9	41.2
0.75	Carbon Emissions	4.1	0.29	99.6	59.7
0.50	Social	0.99	0.14	99.7	47.7
0.50	Environmental	2.4	0.22	99.8	35.7
0.50	Governance	0.59	0.11	99.9	41.2
0.50	Carbon Emissions	1.8	0.19	99.6	59.7
0.25	Social	0.25	0.071	99.7	47.7
0.25	Environmental	0.60	0.11	99.8	35.7
0.25	Governance	0.15	0.054	99.9	41.2
0.25	Carbon Emissions	0.46	0.096	99.6	59.7

Table OA.21: **Score-noise sensitivity.** The table attenuates the observed outcome-intensity direction by reliability ρ_g, holding the estimated supply-and-demand system and Aggregate-Supply CE specification fixed. Portfolio impact return at 100 bps is the model-implied expected outcome change at a CE-return cost of 100 basis points. Direct recovery and Net-outcome recovery are signed equal-CE recovery percentages for the direct tilt and net-outcome tilt.

Recovery across active sets. Table OA.22 summarizes signed equal-CE recovery across full cross-section and restricted active sets.

Observed direct-tilt recovery is generally high across the reported active sets, with low-recovery exceptions in the narrowest SIC3 cases, while net-outcome recovery is more variable. Constructed wrong-direction cases are reported in Figure 5, Table 4, and Section OA.7.

Long-only and divestment comparisons. Table OA.23 compares fixed-investment long-only frontiers and tilts with the unconstrained cases and reports a low-net-outcome divestment path.

Setting	Sets	Cases	Direct recovery	Net-outcome recovery	Corr.	IQR
Market	1	4	99.6 to 99.9	35.7 to 59.7	0.78	0.38
SIC1	7	28	99.3 to 100.0	30.5 to 85.8	0.81	0.37
SIC2	41	164	82.1 to 100.0	17.2 to 99.6	0.83	0.27
SIC3	71	284	19.8 to 100.0	12.8 to 99.8	0.86	0.25

Table OA.22: **Recovery across active sets.** The table summarizes observed signed equal-CE recovery across Market, SIC1, SIC2, and SIC3 active sets in the 2019 frontier sample. Market is unthresholded; SIC1, SIC2, and SIC3 rows include active sets with at least five issuers after the frontier-sample and score-availability filters for the four observed outcome scores: Social, Environmental, Governance, and Carbon Emissions. Market rows use Aggregate-Supply CE; SIC rows use Market-Risk CE with the baseline contribution multiplier matrix. Direct recovery and Net-outcome recovery columns report signed equal-CE recovery percentage ranges. Corr. is the median within-set correlation between the raw score and the impact intensity, and IQR is the median within-set interquartile range of the direct contribution multiplier.

Outcome	Frontier return	Direct	Net-outcome	Long-only frontier	Long-only direct	Long-only net-outcome	Divestment path
Social	0.28	1.00	0.48	0.78	0.77	0.46	0.48
Environmental	0.44	1.00	0.36	0.96	0.96	0.32	0.53
Governance	0.22	1.00	0.41	0.85	0.85	0.32	0.56
Carbon Emissions	0.38	1.00	0.60	0.82	0.81	0.59	0.55

Table OA.23: **Fixed-investment long-only and divestment comparisons.** The table reports the unconstrained Impact Frontier portfolio impact return at 100 basis points of CE-return cost and equal-CE recovery for full cross-section direct and net-outcome tilts, fixed-investment long-only frontiers, fixed-investment long-only direct and net-outcome tilts, and a low-net-outcome divestment path in the full cross-section frontier sample. The long-only columns test whether the direct-versus-net recovery comparison depends on unconstrained short positions. The divestment column is a heuristic exclusion comparison, not a frontier.

Concentration and influence. Table OA.24 tests whether the 2019 frontier is driven by a few issuers or sectors. The optimized frontier is not concentrated in a few issuers: top-ten gross active-weight shares are 32.4 to 35.8 percent and effective issuer counts are 54 to 64 . Dropping the 25 largest issuers changes direct recovery by at most 0.4 percentage points. Sector exclusions lower frontier strength by up to 33 percent and move net-outcome recovery by as much as 9.8 percentage points, while direct recovery changes by at most 0.6 percentage points.

Panel A. Optimized frontier concentration

Outcome	Top 10 $\lvert dV \rvert$	Effective issuers, dV	Top 10 $\lvert dK \rvert$	Effective issuers, dK
Social	35.8	56	33.6	61
Environmental	33.2	54	36.3	48
Governance	32.4	64	32.4	64
Carbon Emissions	33.5	60	31.4	64

Panel B. Influence of large issuers and sectors

Outcome	Drop top 25			Worst SIC1 drop		
	η^2 ratio	Direct rec.	Net rec.	η^2 ratio	Direct rec.	Net rec.
Social	0.80	-0.4	1.3	0.67	0.3	-9.8
Environmental	0.67	-0.3	0.6	0.70	-0.6	-1.0
Governance	0.83	-0.2	2.3	0.76	0.0	-5.2
Carbon Emissions	0.74	0.0	0.7	0.70	0.3	2.0

Table OA.24: **Concentration and influence of the optimized frontier.** Panel A reports issuer concentration for the Impact Frontier at the common 100 basis point CE-return cost. Panel B recomputes frontier strength and recovery after removing the 25 largest issuers or the most influential one-digit SIC sector. dV is the active dollar tilt and dK is the induced equilibrium capital response. Top shares use gross absolute values, effective issuers are inverse Herfindahl counts, and recovery changes are percentage-point changes relative to the full cross-section baseline for the same outcome.

Frontier portfolio and capital response. Figure OA.1 reports the firm-by-firm quantities behind the frontier portfolio construction and separates the investor's active portfolio direction from the equilibrium capital response.

Issuer size and direct contribution multipliers. Figure OA.2 shows why equal-weight and market-value-weighted summaries differ. Direct contribution multipliers rise with issuer

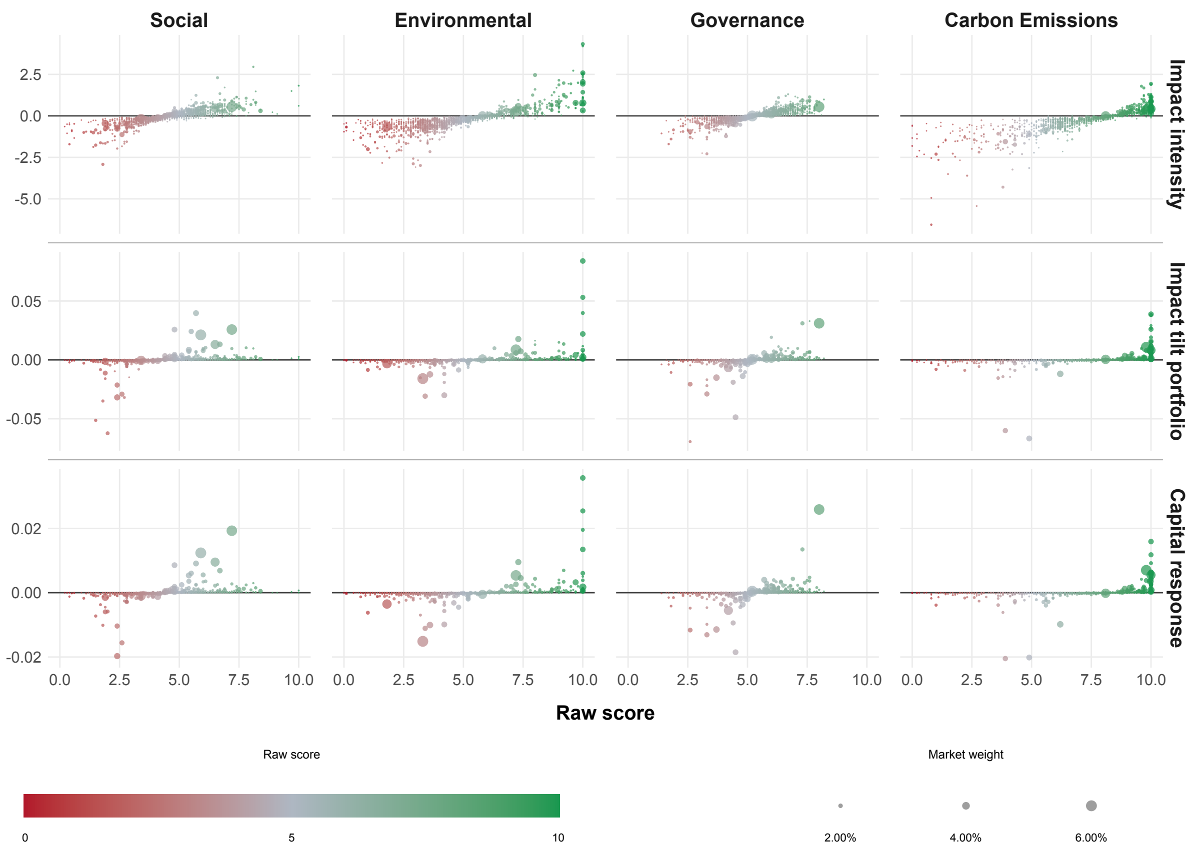

Figure OA.1: **Raw scores, frontier tilts, and capital responses.** Each column corresponds to one outcome category. The horizontal axis is the raw score. Rows report the impact intensity, the Impact Frontier portfolio-weight tilt normalized to one unit of gross active weight, and the induced capital response $\boldsymbol{C}\,\delta\boldsymbol{k}_i$. Point size is market portfolio weight, and color repeats the raw score. Pearson correlations with the raw score for Social, Environmental, Governance, and Carbon Emissions are: impact intensity (0.79, 0.78, 0.80, 0.78), impact tilt portfolio (0.28, 0.29, 0.28, 0.30), and capital response (0.28, 0.28, 0.27, 0.30).

size in this public-equity calibration because issuer-demand slopes fall relative to investor-supply slopes. The non-market-risk term raises baseline multipliers but does not overturn this size ordering.

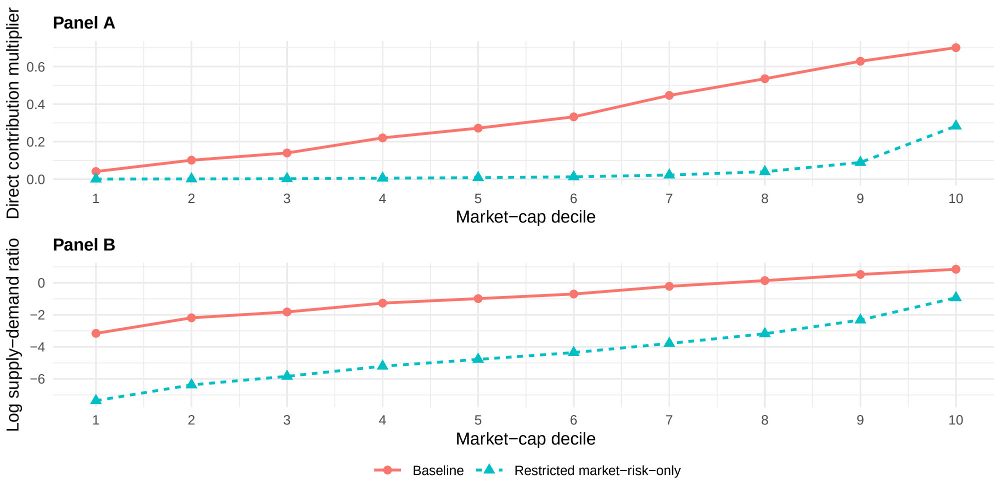

Figure OA.2: **Issuer size and direct contribution multipliers.** Panel A reports median direct contribution multipliers by market-cap decile under the baseline and restricted market-risk-only specifications. Panel B reports the median log supply-demand ratio, $\theta_n = \log \Phi_{S,nn} - \log \Phi_{D,nn}$, for the same deciles. Because $\mathcal{C}_n^{\text{dir}} = (1 + \exp(-\theta_n))^{-1}$, higher values of θ_n correspond to higher direct contribution multipliers.

OA.7 Analytical recovery calculations

This section characterizes the constructed worst-recovery cases in Figure 5. The calculation uses the same outcome-intensity space as the figure: after applying the baseline risk-free-asset netting, candidate directions have weighted mean zero and weighted standard deviation one under the market-portfolio weights used for the corresponding active-set calculation. Fix an active opportunity set $\mathcal{A}$, and embed the inverse effective covariance matrix in the full asset space as

$$\Xi_{\mathcal{A}} = \boldsymbol{I}'_{\mathcal{A}} \left(\boldsymbol{\Sigma}^{\text{eff}}_{\mathcal{A}}\right)^{-1} \boldsymbol{I}_{\mathcal{A}}. \tag{OA.29}$$

Let $\widetilde{\boldsymbol{\mathcal{C}}}^M$ denote the comparison matrix. The net-outcome tilt uses $\widetilde{\boldsymbol{\mathcal{C}}}^I = \boldsymbol{I}$, while the direct tilt uses $\widetilde{\boldsymbol{\mathcal{C}}}^D = \operatorname{diag}(\boldsymbol{\mathcal{C}}^{\text{dir}})$. Define

$$\boldsymbol{H}_{\mathcal{A}}^{CC} = \boldsymbol{\mathcal{C}}\Xi_{\mathcal{A}}\boldsymbol{\mathcal{C}}', \qquad \boldsymbol{H}_{\mathcal{A}}^{MM} = \widetilde{\boldsymbol{\mathcal{C}}}^M \Xi_{\mathcal{A}} \widetilde{\boldsymbol{\mathcal{C}}}^{M\prime}, \tag{OA.30}$$

$$\boldsymbol{\Psi}_{\mathcal{A}}^M = \frac{1}{2}\left(\boldsymbol{\mathcal{C}}\Xi_{\mathcal{A}}\widetilde{\boldsymbol{\mathcal{C}}}^{M\prime} + \widetilde{\boldsymbol{\mathcal{C}}}^M \Xi_{\mathcal{A}}\boldsymbol{\mathcal{C}}'\right). \tag{OA.31}$$

For a candidate vector of net outcome intensities $\boldsymbol{g}$, signed equal-CE recovery is

$$\rho_{\mathcal{A},M}(\boldsymbol{g}) = \frac{\boldsymbol{g}'\boldsymbol{\Psi}_{\mathcal{A}}^M\boldsymbol{g}}{\left[\left(\boldsymbol{g}'\boldsymbol{H}_{\mathcal{A}}^{CC}\boldsymbol{g}\right)\left(\boldsymbol{g}'\boldsymbol{H}_{\mathcal{A}}^{MM}\boldsymbol{g}\right)\right]^{1/2}}. \tag{OA.32}$$

The constructed worst-recovery distribution minimizes (OA.32) over the centered and scaled outcome-intensity space used in Figure 5. Because $\rho_{\mathcal{A},M}(\boldsymbol{g})$ is homogeneous of degree zero, the weighted-standard-deviation constraint fixes the reported score scale; the weighted-mean-zero restriction fixes the admissible direction space.

Let $\boldsymbol{Q}_{\mathcal{A}}$ have columns spanning the admissible outcome-intensity subspace, with weighted mean equal to zero, and write $\boldsymbol{g} = \boldsymbol{Q}_{\mathcal{A}}\boldsymbol{z}$. Define

$$\bar{\boldsymbol{H}}_{\mathcal{A}}^{CC} = \boldsymbol{Q}_{\mathcal{A}}'\boldsymbol{H}_{\mathcal{A}}^{CC}\boldsymbol{Q}_{\mathcal{A}}, \qquad \bar{\boldsymbol{H}}_{\mathcal{A}}^{MM} = \boldsymbol{Q}_{\mathcal{A}}'\boldsymbol{H}_{\mathcal{A}}^{MM}\boldsymbol{Q}_{\mathcal{A}}, \qquad \bar{\boldsymbol{\Psi}}_{\mathcal{A}}^{M} = \boldsymbol{Q}_{\mathcal{A}}'\boldsymbol{\Psi}_{\mathcal{A}}^{M}\boldsymbol{Q}_{\mathcal{A}}.$$

For $\mu(\boldsymbol{z}) = \left(\frac{\boldsymbol{z}'\bar{\boldsymbol{H}}_{\mathcal{A}}^{MM}\boldsymbol{z}}{\boldsymbol{z}'\bar{\boldsymbol{H}}_{\mathcal{A}}^{CC}\boldsymbol{z}}\right)^{1/2}$, stationary directions within the admissible subspace satisfy

$$2\bar{\boldsymbol{\Psi}}_{\mathcal{A}}^M\boldsymbol{z} = \rho\left(\mu\bar{\boldsymbol{H}}_{\mathcal{A}}^{CC} + \mu^{-1}\bar{\boldsymbol{H}}_{\mathcal{A}}^{MM}\right)\boldsymbol{z}. \tag{OA.33}$$

The constructed worst cases in Figure 5 solve this restricted problem numerically, using fixed-μ generalized eigenvectors and the observed outcome scores as starts, then minimizing the true signed-recovery ratio. The formulas below describe why the resulting minima can be far from one; the unrestricted formulas are diagnostic benchmarks, not replacements for the centered search used in Figure 5.

Full cross-section net-outcome tilt under Aggregate-Supply CE. Under the full cross-section Aggregate-Supply CE specification, $\boldsymbol{\Xi}$ is proportional to $\boldsymbol{\Phi}_S^{-1}$. Define the symmetric positive-definite response matrix $\boldsymbol{\Phi}_R \equiv \boldsymbol{\Phi}_S^{1/2}(\boldsymbol{\Phi}_T)^{-1}\boldsymbol{\Phi}_S^{1/2}$. Its eigenvalues are the full cross-section Aggregate-Supply CE modal contribution multipliers. Setting $\boldsymbol{y} = \boldsymbol{\Phi}_S^{-1/2}\boldsymbol{g}$, net-outcome recovery becomes

$$\rho_I(\boldsymbol{y}) = \frac{\boldsymbol{y}'\boldsymbol{\Phi}_R\boldsymbol{y}}{[(\boldsymbol{y}'\boldsymbol{\Phi}_R^2\boldsymbol{y})(\boldsymbol{y}'\boldsymbol{y})]^{1/2}}. \tag{OA.34}$$

Over the unrestricted nonzero $\boldsymbol{y}$-space, the first antieigenvalue of $\boldsymbol{\Phi}_R$ implies

$$\min_{\boldsymbol{y}\neq 0} \rho_I(\boldsymbol{y}) = \frac{2\sqrt{\lambda_{\min}(\boldsymbol{\Phi}_R)\lambda_{\max}(\boldsymbol{\Phi}_R)}}{\lambda_{\min}(\boldsymbol{\Phi}_R) + \lambda_{\max}(\boldsymbol{\Phi}_R)}. \tag{OA.35}$$

The 2019 calibration has $\lambda_{\min}(\boldsymbol{\Phi}_R) = 0.0022$ and $\lambda_{\max}(\boldsymbol{\Phi}_R) = 0.93$, which imply the unrestricted minimum 0.097. The numerical search reports 0.097, so the two values agree at the displayed precision in this calibration.

An unrestricted diagonal proxy replaces the extreme modal contribution multipliers with the smallest and largest direct contribution multipliers. If these are $d_{\min}$ and $d_{\max}$, then

$$\rho_{I,\text{diag}} = \frac{2\sqrt{d_{\min}d_{\max}}}{d_{\min} + d_{\max}}. \tag{OA.36}$$

The empirical values $d_{\min} = 0.0022$ and $d_{\max} = 0.93$ imply 0.097. Thus contribution-multiplier dispersion reproduces both the unrestricted antieigenvalue calculation and the numerical worst-case search almost exactly.

Many-asset direct tilt scale calculation. The following approximation isolates the accumulation of small cross-issuer responses. Let $\mathbf{1}_n \in \mathbb{R}^n$ denote the vector of ones. The approximation uses the same CE cost in every direction and replaces the empirical matrices with a diagonal term and a common off-diagonal term:

$$\boldsymbol{\mathcal{C}}_{\text{cs}} = (d-m)\boldsymbol{I}_n + m\mathbf{1}_n\mathbf{1}_n', \qquad \widetilde{\boldsymbol{\mathcal{C}}}_{\text{cs}}^D = d\boldsymbol{I}_n, \qquad \boldsymbol{\Xi}_{\text{cs}} = \boldsymbol{I}_n, \qquad r = \frac{m}{d}, \tag{OA.37}$$

with $0 \leq r < 1$.

If outcome-intensity directions are unrestricted, the common response mode has relative multiplier $1 + (n-1)r$, while every orthogonal mode has relative multiplier $1 - r$. The unrestricted worst-case direct-tilt recovery in this approximation is

$$\rho_{D,\min}^{\text{unres}}(n,r) = \frac{2\sqrt{[1+(n-1)r](1-r)}}{2+(n-2)r}. \tag{OA.38}$$

Figure 5 and Table 4 do not use unrestricted outcome-intensity directions. The constructed directions are market-weighted-mean-zero after baseline risk-free-asset netting and are normalized to market-weighted unit standard deviation. The unit-standard-deviation restriction fixes scale and does not affect signed recovery. The market-weighted-mean-zero restriction changes the feasible directions.

Let $\boldsymbol{w}$ be the market-weight vector used in the constructed-direction calculation, normalized so that $\boldsymbol{w}'\mathbf{1}_n = 1$. Let

$$\boldsymbol{P}_w \equiv \boldsymbol{I}_n - \boldsymbol{w}(\boldsymbol{w}'\boldsymbol{w})^{-1}\boldsymbol{w}', \qquad s_w \equiv \|\boldsymbol{P}_w \mathbf{1}_n\|^2 = n - \frac{(\boldsymbol{w}'\mathbf{1}_n)^2}{\boldsymbol{w}'\boldsymbol{w}} = n - \frac{1}{\boldsymbol{w}'\boldsymbol{w}}. \tag{OA.39}$$

The scalar s_w is the squared length of the common response direction after projection into the market-weighted-mean-zero subspace. Within the common-response approximation, the restricted feasible subspace has relative multipliers $1 - r$ and $1 + (s_w - 1)r$. This yields the restricted common-response approximation to worst-case direct-tilt recovery:

$$\rho_{D,\min}^{\text{res}}(s_w,r) = \frac{2\sqrt{[1+(s_w-1)r](1-r)}}{2+(s_w-2)r} = \frac{2\sqrt{1+c_w}}{2+c_w}, \tag{OA.40}$$

with $c_w = s_w r/(1-r)$.

The unrestricted formula is the special case $s_w = n$, so $c_w = nr/(1-r)$. With equal weights, the mean-zero restriction removes the common response mode, $s_w = 0$, and the common-response approximation has recovery one. With market weights, the projection usually leaves part of the common response mode in the feasible subspace, so $0 < s_w < n$. For Figure 5, the relevant approximation is equation (OA.40), using the same market weights as the constructed-direction calculation. For the full cross-section calculation, $n = 1200$, $s_w = 1100$, and $r = 0.0011$. Equation (OA.40) then implies recovery of 0.92, close to the full-matrix constructed value of 0.94.

Restricted active sets under Market-Risk CE. For a restricted active set $\mathcal{A}$, Market-Risk CE uses the active-set restriction of $\mathbf{\Phi}_S^M$ to measure financial cost, while the contribution multiplier matrix remains $\boldsymbol{\mathcal{C}} = (\mathbf{\Phi}_S^M + \mathbf{\Phi}_S^N + \mathbf{\Phi}_D)^{-1}(\mathbf{\Phi}_S^M + \mathbf{\Phi}_S^N)$. Thus $\mathbf{\Xi}_{\mathcal{A}}^M = \boldsymbol{I}'_{\mathcal{A}} \left(\boldsymbol{I}_{\mathcal{A}} \mathbf{\Phi}_S^M \boldsymbol{I}'_{\mathcal{A}}\right)^{-1} \boldsymbol{I}_{\mathcal{A}}$, up to the common CE scaling. For the net-outcome tilt, the signed-recovery numerator is governed by

$$\mathbf{\Psi}_{\mathcal{A}}^{I,M} = \frac{1}{2}\left(\boldsymbol{\mathcal{C}}\mathbf{\Xi}_{\mathcal{A}}^M + \mathbf{\Xi}_{\mathcal{A}}^M \boldsymbol{\mathcal{C}}'\right). \tag{OA.41}$$

Unlike the full cross-section Aggregate-Supply CE case, the full-space matrix in (OA.41) need not be positive semidefinite. A negative full-space eigenvalue identifies a candidate wrong-direction mode, subject to the financing-neutral score restriction.

All 41 SIC2 active sets used in the 2019 restricted recovery analysis contain exactly 1 negative full-space numerator mode under Market-Risk CE. The constrained search confirms that admissible wrong-direction recovery is possible: the constructed SIC2 59 direction reaches recovery of -1, while its Aggregate-Supply CE comparison is positive at approximately 0.32. Recovery remains positive in all 164 observed outcome-by-SIC2 cases.

OA.8 Blended score and outcome effectiveness

The outcome-effectiveness matrix interprets a values vector through the CE-return-cost calculation for the capital-allocation channel. Let $\boldsymbol{R}_t$ collect the aggregate-supply impact-return vectors for the four outcome categories in market-value-position units, and let $\boldsymbol{\nu}$ denote a values vector over those outcomes. The composite impact return is $\boldsymbol{R}_t\boldsymbol{\nu}$. Using the same estimated effective covariance matrix $\hat{\mathbf{\Sigma}}_{S,t}^{\text{eff}} = V_{M,t}\hat{\mathbf{\Phi}}_{S,t}^V$ to compute CE-return costs, the outcome-effectiveness matrix is

$$\mathbf{M}_t \equiv \boldsymbol{R}'_t(\hat{\mathbf{\Sigma}}_{S,t}^{\text{eff}})^{-1}\boldsymbol{R}_t.$$

This matrix summarizes the CE-return-cost-weighted effectiveness of the outcome categories. For a fixed values vector $\boldsymbol{\nu}$, the vector $\mathbf{M}_t\boldsymbol{\nu}$ indicates which outcome categories the capital-allocation channel can move most effectively at the margin under the same CE-return-cost calculation. It is not a replacement for the values vector; it is the channel's marginal conversion of those stated values into outcome movements, showing which valued outcomes are cheap or expensive to move through capital allocation.